Exposed: Shedding *Blacklight* on Online Privacy*

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Abstract

Who is the most surveilled online? One account holds that risk follows vulnerability, so the least digitally literate should be the most exposed. A competing account holds that risk follows opportunity, so exposure should track how much time a person spends online. We pair one month of passively observed, anonymized browsing by 1,134 US adults with domain-level Blacklight audits of seven tracking techniques. The panelists logged 4.4 million visits to 64,074 domains. The audits reached domains carrying 79% of those visits. The domains they missed were mostly live sites that blocked the scanner and we count every visit to them as carrying no trackers at all. Every estimate is therefore a floor. Tracking is near universal and immediate: over 99% of users encounter ad trackers and third-party cookies, half of them within twelve hours, and session recording, keylogging, and canvas fingerprinting reach 43 to 53% of users within two days. For the median user, the biggest observer of their browsing sees 58% of it. For nine users in ten, that observer is Google. Risk follows opportunity for education: the better educated meet more trackers over the month, and the gap closes once we hold browsing volume constant. It does not for age: users 65 and older encounter 76 to 87% more trackers per visit than users under 25. Where people browse, and not only how much, is associated with the surveillance they meet.

Keywords: Online Privacy, Online Safety, Digital Divide, KeyLogging, Tracking

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*Replication materials: https://github.com/themains/private_blacklight

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1 Introduction

Two theories predict who is most at risk online. The first is the digital divide: risk follows vulnerability, so women, older people, racial minorities, and the less educated should be the most exposed, being less equipped to recognize what threatens them (van Deursen and van Dijk, 2011; Hadlington and Chivers, 2020; Whitty, 2019). The second is exposure through presence: risk follows opportunity, so whoever spends the most time online meets the most of whatever is out there, whatever their skill (Simoiu et al., 2020; Baki and Verma, 2023). Where the two have been tested against observed behavior rather than self-report, the second has largely prevailed. Demographic differences in exposure to malicious domains vanish at the median once online presence is accounted for (Shen and Sood, 2025), and the people most often caught up in data breaches are the better educated and the heavily online (Cor and Sood, 2018).

Web tracking should not behave the same way. Malware can be recognized and evaded, and people do evade it, leaving quickly once they land on a bad site (Shen and Sood, 2025). Breached accounts follow from accounts a person chose to open. Tracking is structurally different. It is installed by the sites a person visits, under arrangements between those sites and third parties the visitor has no part in choosing, and it runs whether or not the visitor knows it is there.[1] Skill and caution should therefore have less purchase on tracking than on malware or on breached credentials. If that is right, the composition of a person's browsing, and not merely its volume, should govern how much surveillance they meet.

Testing the claim means measuring how much tracking people meet, not how much tracking sites carry. The literature has mostly measured the latter (Englehardt and Narayanan, 2016; Karaj et al., 2019). Such crawls find tracking nearly everywhere. But people do not visit sites at random (Dambra et al., 2022). The studies that do start from real browsing data carry no demographics, and so cannot say who is exposed (Yu et al., 2016; Dambra et al., 2022). We start from a representative sample of the US adult population, and we know the age, gender, race, and education of everyone in it. Our estimand is the rate at which a person's browsing meets tracking technology, and the share of that browsing any single organization can observe.

We ask four questions:

[1] Consent interfaces qualify this without overturning it. They were less common in the United States than in Europe during our observation window (**?**), and where they appear they solicit agreement to purposes rather than to named vendors, of which the IAB Global Vendor List now registers several thousand. Audits also find tracking firing before any choice is made and after consent is refused (**??**). The only US statute operative in June 2022 was the CCPA, whose minimum disclosure requirement went widely unmet (**?**).

- **RQ1:** How much tracking, and of which kinds, does a person meet over a month of ordinary browsing, and how soon after they start?
- **RQ2:** How much of one person's browsing can a single organization observe, and which organization is it?
- **RQ3:** How does exposure differ by age, gender, race, and education, before and after holding browsing volume constant?
- **RQ4:** Does surveillance fall on the same people as the online risks it is usually grouped with?

To answer them, we pair one month of passively metered browsing, 4.4 million visits to 64,074 domains by 1,134 adults on a YouGov panel matched to US population margins, with audits of every visited domain by Blacklight (Mattu and Sankin, 2020), which detects seven techniques from ad trackers and third-party cookies through session recording, keylogging, and canvas fingerprinting. Linking the detected third parties to their parent organizations through DuckDuckGo's Tracker Radar converts encounters into measures of surveillance by organizations.

We find tracking is near universal and immediate. More than 99% of users encounter ad trackers and third-party cookies, half of them inside twelve hours of start of measurement window, and invasive techniques (session recording, keylogging, and canvas fingerprinting) reach 42 to 52% of users within two days. For the median user, the biggest observer of their browsing sees 58% of it, and for nine users in ten that observer is Google. Because nearly everyone is tracked, who is most exposed is a question of degree. Risk follows opportunity for education, as it does for other online risks: the better educated meet more trackers over the month, and the gap closes once browsing volume is held constant. It does not for age. Users 65 and older meet 76 to 87% more trackers per visit than users under 25, the gradient runs smoothly across birth cohorts rather than resting on where the bins were cut, and the gap survives both ways of holding volume constant. Demographics taken together, however, still account for little of the total variance in exposure, so the inequality in surveillance lies overwhelmingly within demographic groups rather than between them.

Our main contributions can be summarized as follows:

- We assemble the first dataset joining passively metered browsing, participant demographics, and domain-level audits of seven tracking techniques, on a sample built to represent US adults.
- We measure exposure at the level of individual people rather than sites, including per-user concentration across the organizations that operate the trackers.
- We show that the explanation that accounts for demographic gaps in other online risks, that they reflect time spent online, does not extend to surveillance, and that the two kinds of risk fall on opposite ends of the age distribution.
- We demonstrate a way to validate studies that pair browsing data with later audits, using historical crawl archives to measure how much the timing of an audit, and its failures, shift the resulting estimates (Section 3.6).
- We release the anonymized browsing panel (Sood, 2022), the full audit results, and the linkage and analysis code.

2 Related Work

2.1 Tracking Measured from Sites

Almost everything known about web tracking has been learned via websites rather than browsing behavior. The canonical study, Englehardt and Narayanan (2016), audited a million sites and found stateful and stateless tracking throughout. Parallel work established that fingerprinting was already deployed in the wild (Acar et al., 2013; Nikiforakis et al., 2013; Acar et al., 2014; Mowery and Shacham, 2012), that it could be recognized from a script's behavior rather than from a blocklist (Iqbal et al., 2021), and that forms leak email addresses before anyone presses submit (Senol et al., 2022). Others traced where the collected data travels once it leaves the page, mapping the cookie ecosystem and the parties operating in it (Sanchez-Rola and Santos, 2018; Sanchez-Rola et al., 2021).

The same methods have since been turned on how long tracking persists and where it matters most. Standing observatories follow the ecosystem month by month (Karaj et al., 2019), and longitudinal studies follow it across decades (Lerner et al., 2016; Solomos et al., 2020). Targeted audits find it on hospital, pharmacy, and government health sites, where the stakes run highest (Niforatos et al., 2021; Zheutlin et al., 2022b,a). Blacklight, the tool we use, belongs to this tradition (Mattu and Sankin, 2020).

Crawls have also settled a second question, which is whether users can decline any of this. Consent interfaces are less common in the United States than in Europe, and trackers multiply once a banner is accepted (?). Where banners do appear, audits find widespread non-compliance: tracking that fires before any choice is made, and tracking that continues after consent is refused (??). In the United States, adoption of even the CCPA's minimum disclosure requirement remains low (?). Whatever users are asked, then, is a poor guide to what they get.

The inventory this work has built is detailed and, for its purpose, close to complete, but it describes the web rather than the people who browse it. A crawl counts each site once, and no one visits the web the way a crawler does, so we cannot use these studies to understand how much tracking any particular person meets.

2.2 Tracking Measured from Browsing

A second strand starts from browsing rather than from sites, but not from representative samples. Yu et al. (2016) measured tracker reach across 21 million page loads from 200,000 users and found a single organization present on 42% of page visits in Germany. Dambra et al. (2022) joined browsing telemetry from 250,000 machines to custom crawls and showed that exposure measured from the user's side is more concentrated than exposure measured from the tracker's. Both are large convenience samples. The first drew its users from Cliqz, a privacy-focused browser, and so selected on the disposition under study. The second drew its users from the customers of an antivirus vendor, all of them on Windows. Neither study has demographics. Smaller studies inherit the same limits, with samples drawn from a few tens of volunteers enrolled through a Chrome extension (Hu et al., 2020), some 1,300 volunteers recruited through university mailing lists and researchers' personal contacts (Pugliese et al., 2020), and 30 paid crowdworkers sent to an assigned list of 100 sites (Muthu Selva Annamalai et al., 2025). To our knowledge, no study has measured tracking exposure from the browsing of a sample built to represent a population.

2.3 Who Bears Online Risk

Little is known about disparities in online tracking. Berke et al. (2025) collected browser fingerprints and demographics from 8,400 US participants and found lower-income and older users at greater risk of fingerprinting. This study is close in spirit to ours, but measures something distinct. Device identifiability is about hardware and device settings. Exposure, our measure, is a property of where people go. The distinction carries practical weight, because a browser vendor can change the first, while the second changes only when the sites people visit change.

Other online harms, such as malware, phishing, and breached credentials, have a fuller user-centric literature. Its long-standing expectation is that risk follows vulnerability, so the less digitally skilled should be the most exposed (van Deursen and van Dijk, 2011; Hadlington and Chivers, 2020; Whitty, 2019). That expectation rests on measures of skill, which is unevenly distributed by education and by age (van Deursen and van Dijk, 2011), rather than on measures of exposure. Where exposure itself has been observed, the finding is different: it follows opportunity, and demographic gaps shrink or invert once time spent online is taken into account (Simoiu et al., 2020; Baki and Verma, 2023; Shen and Sood, 2025; Cor and Sood, 2018).

Whether web tracking follows the same pattern has not been tested. And there are grounds for expecting that it does not. Malware can be recognized and avoided, and breached credentials follow from accounts a person chose to open, so skill and caution have some purchase on both. Tracking, on the other hand, is installed by the sites a person visits, by parties the visitor never selected, and operates whether or not the visitor is aware of it.

We test whether that difference shows up in who is exposed.

3 Research Design, Data, and Measures

We combine three data sources. The first is a month of passively collected, anonymized, domain-level browsing from a YouGov panel of 1,200 US adults, sampled to match the US adult population and benchmarked against the Current Population Survey, covering 4.4 million visits (Section 3.1). The second is domain-level audits from Blacklight, a scanning tool built by The Markup that detects seven tracking technologies (Section 3.2). The third is the DuckDuckGo Tracker Radar, an open-source list mapping tracker domains to their parent organizations (Section 3.4). Joining the three yields, for each person, the tracking technology present on every domain they visited and the share of their browsing visible to any one organization.

From the join we build two measures of exposure (Section 3.3). A person can meet a lot of trackers because they browse a great deal, or because the sites they visit carry more tracking. Cumulative exposure, the count of tracker encounters over the month, reflects both. The exposure rate, that count divided by the person's visits, isolates the second. Dividing by visits is one way to hold browsing volume constant; Section 3.5 reports a second, the specification used in studies of exposure to other online risks (Shen and Sood, 2025). Two features of the join push the estimates down. Blacklight completed scans for 34,512 of the 64,074 unique domains (53.9%), covering 79.0% of visits, and visits to unscanned domains contribute zero trackers. Separately, the browsing predates the scans by thirty-one months. Using HTTP Archive, the Wayback Machine, and fresh Blacklight rescans, we show that both features bias exposure downward, and that neither moves the demographic comparisons (Section 3.6).

Blacklight could not scan every domain, and it scanned long after the browsing took place. It completed scans for 34,512 of the 64,074 unique domains (53.9%), covering 79.0%

of visits; visits to the rest count as zero trackers. The browsing itself predates the scans by thirty-one months. Both gaps make our estimates too low rather than too high, and neither moves the demographic comparisons, as we show with HTTP Archive, the Wayback Machine, and fresh Blacklight rescans (Section 3.6).

3.1 Browsing Data

Our browsing data comes from YouGov, which draws a random sample from a large synthetic sampling frame (Rivers and Bailey, 2009), invites the matching panelists to participate, and substitutes similar individuals for non-respondents. Our sample consists of 1,200 American adults who volunteered to install a passive metering application, YouGov Pulse, developed with RealityMine, on their devices in exchange for rewards. The application collected de-identified browsing over one month in June 2022 (Shen and Sood, 2025; Sood, 2022; Sood and Shen, 2024), logging domain visits with anonymized URLs (for example, *https://www.google.com/search?ANONYMIZED*) and timestamps, regardless of browser type or privacy settings. All participants gave informed consent and could opt out at any time. Passwords and secure form entries were never collected, and all data were anonymized (Section A).

Sixty-five panelists recorded no online activity during the month and one recorded visits without usable metadata; we exclude all sixty-six. The analytical sample is the remaining 1,134 panelists, who made 4.4 million visits to 64,074 unique domains over the month.

The sample tracks the Current Population Survey (CPS) closely but not perfectly. Gender, education, and age margins are statistically indistinguishable from the CPS; race is not (χ^2 tests in Table 1), and the rejection is driven largely by the small "Other" category, whose definition differs between YouGov and the CPS. Across the sixteen demographic margins in Table 1, the mean absolute deviation from the weighted CPS 2022 shares is 1.3 percentage points and the maximum is 2.6 (high school diploma or below).

Table 1. Sample description and benchmark against the CPS

	YouGov 2022 (N = 1,134)		CPS 2022 (N = 114,158)	
	n	Share	Share	Δ (pp)
Female	595	52.5%	51.2%	+1.3
Male	539	47.5%	48.8%	−1.3
White	720	63.5%	62.1%	+1.4
Hispanic	168	14.8%	17.2%	−2.4
Black	144	12.7%	12.0%	+0.7
Other	56	4.9%	2.5%	+2.4
Asian	46	4.1%	6.2%	−2.1
High school diploma or below	411	36.2%	38.8%	−2.6
Some college	326	28.7%	26.4%	+2.3
College graduate	255	22.5%	22.1%	+0.4
Postgraduate	142	12.5%	12.7%	−0.2
< 25 years old	106	9.3%	11.4%	−2.1
25–34 years old	200	17.6%	17.5%	+0.1
35–49 years old	291	25.7%	24.5%	+1.2
50–64 years old	280	24.7%	24.6%	+0.1
65+ years old	257	22.7%	22.0%	+0.7

Note: CPS shares are weighted Annual Social and Economic Supplement 2022 margins for adults 18 and over under the same category definitions. The "Other" race category bundles groups differently than the CPS recode. χ^2 goodness-of-fit tests of the sample against the CPS shares: gender $\chi^2(1) = 0.7$, $p = .40$; race $\chi^2(4) = 40.5$, $p < .001$; education $\chi^2(3) = 4.3$, $p = .24$; age $\chi^2(4) = 4.8$, $p = .30$. Mean absolute deviation across the 16 categories is 1.3 percentage points; the maximum is 2.6 (High school diploma or below).

3.2 Measuring Tracking on Domains

Blacklight is an on-demand privacy inspection tool that simulates a fresh user arriving at a website and scans for seven stateful and stateless tracking methods, identifying them through browser automation, network request monitoring, and behavioral script analysis (Mattu and Sankin, 2020). Scans ran in a headless Chromium 126 browser emulating an iPhone 13 Mini user agent from a US-based environment, with a fresh browser cache per scan, a 30-second per-page timeout, and coverage of the homepage plus one randomly selected internal page (Section B). The emulated device makes little difference to how much tracking a crawl finds: HTTP Archive, which crawls the same domains as both a phone and a desktop browser, reports nearly identical ad-tracker prevalence on the two in 2022 (94.0% against 94.1%) and in 2025 (91.8% against 91.9%) (Table C.6). We submitted the 64,074 unique domains visited in our sample to Blacklight in January 2025 and obtained results for 34,512 (53.9%), covering

79.0% of all visits. Section 3.6 shows that the scan timing and the unscanned visits both make our estimates conservative.

Blacklight detects the following seven methods, of which session recording, keylogging, and canvas fingerprinting are the most invasive (Acar et al., 2014; Karaj et al., 2019; Mattu and Sankin, 2020; Mowery and Shacham, 2012; Senol et al., 2022; Berke et al., 2025); Section B gives the details.

- **Ad trackers:** outgoing requests matched to DuckDuckGo's "Ad Motivated Tracking" list.
- **Third-party cookies:** `Set-Cookie` headers on requests to third-party services.
- **Facebook Pixel:** requests to Facebook domains whose URL parameters carry pixel events.
- **Google Analytics (remarketing):** requests to Google domains carrying a Google account identifier. The measure flags the "remarketing audiences" feature, not the presence of Google Analytics.
- **Session recording:** script behavior together with a known list of URLs for session replay services.
- **Keylogging:** known values typed into form fields, then network activity monitored for exfiltration of those exact keystrokes.
- **Canvas fingerprinting:** `<canvas>` behavior and pixel-level script outputs.

3.3 Measuring Exposure to Tracking Methods

Each individual i has a set of site visits $\mathcal{V}_i$, where each visit v is a timestamped instance of visiting a page on domain $d(v)$, and $|\mathcal{V}_i|$ is the individual's total visits in the month. Let $\text{trackers}_d^{(s)}$ be the set of trackers of type s that Blacklight detected on domain d; for domains the scan did not reach, this set is empty by construction. Cumulative exposure aggregates tracker counts over the domains visited (Equation (1)), and the exposure rate normalizes that total by visits to adjust for browsing intensity (Equation (2)).

$$\text{Cumulative Exposure}_i^{(s)} = \sum_{v \in \mathcal{V}_i} \left| \text{trackers}_{d(v)}^{(s)} \right|, \tag{1}$$

$$\text{Exposure Rate}_i^{(s)} = \frac{1}{|\mathcal{V}_i|} \text{Cumulative Exposure}_i^{(s)}. \tag{2}$$

3.4 Measuring Tracking by Organizations

A tracker matters less as a script than as a party. The Blacklight results identify the third-party domains embedded on each visited domain (for example, *connect.facebook.net*), and we link those to parent organizations (*Facebook, Inc.*) using the DuckDuckGo Tracker Radar (https://github.com/duckduckgo/tracker-radar), which maps over 38,000 third-party tracking domains to over 19,000 parent organizations. Let O_{iv} denote the set of parent organizations present on visit v by individual i, counting an organization once however many of its third-party domains appear. This lets us quantify the number of distinct organizations tracking each user (Equation (3)) and the share of a user's browsing visible to any one organization j (Equation (4)).

$$|\text{Organizations}_i| = \left| \bigcup_{v \in \mathcal{V}_i} O_{iv} \right| \tag{3}$$

$$\text{Tracking share}_{ij} = \frac{\sum_{v \in \mathcal{V}_i} \mathbf{1}(j \in O_{iv})}{|\mathcal{V}_i|} \tag{4}$$

We also compute each organization's share weighted by dwell time t_{iv} rather than by visits, and reach similar findings (Section I):

$$\text{Tracking share}_{ij}^{(\text{dur})} = \frac{\sum_{v \in \mathcal{V}_i} \mathbf{1}(j \in O_{iv})\, t_{iv}}{\sum_{v \in \mathcal{V}_i} t_{iv}}. \tag{5}$$

3.5 Demographic Differences

We estimate disparities in two ways, which differ in how they hold browsing volume constant. The first regresses exposure on demographics alone:

$$y_i = \alpha + \beta\,\text{woman}_i + \sum_k \gamma_k\,\text{race}_{ik} + \sum_k \delta_k\,\text{education}_{ik} + \sum_k \lambda_k\,\text{age group}_{ik} + \varepsilon_i, \quad (6)$$

where y_i is individual i's exposure to one of the seven tracking methods and k indexes the non-reference categories of each covariate. We estimate by ordinary least squares with Huber-White standard errors, once on cumulative exposure and once on the exposure rate. Covariates are gender (woman; ref. man), race and ethnicity (African American, Asian, Hispanic, Other; ref. White), education (some college, college degree, postgraduate; ref. high school or less), and age group (25–34, 35–49, 50–64, 65+; ref. 18–24). All predictors are indicators, so coefficients are directly comparable. Volume enters here only through the choice of outcome: the rate divides cumulative exposure by visits, which holds volume constant by imposing proportionality between browsing and exposure.

The second adjustment does not impose proportionality. Following the specification used in studies of exposure to other online risks (Shen and Sood, 2025), we enter online presence directly, as a conditional quantile of cumulative exposure:

$$Q_\tau(y_i \mid X_i) = \alpha + \beta\,\text{woman}_i + \sum_k \gamma_k\,\text{race}_{ik} + \sum_k \delta_k\,\text{education}_{ik} + \sum_k \lambda_k\,\text{age group}_{ik} + \eta_1 \text{visits}_i + \eta_2 \text{visits}_i^2, \quad (7)$$

where X_i collects the covariates and visits_i is individual i's total visits during the month, scaled to the unit interval. We estimate at the median ($\tau = 0.5$) with bootstrapped standard errors, because exposure is heavily right-skewed and the median is the summary that skew does not distort.

The two adjustments make different assumptions. The rate assumes that doubling a person's visits doubles the trackers they meet. That may not hold: heavy browsers return to the same handful of sites, so each additional visit brings less new tracking than the last. If exposure grows that way, dividing by visits penalizes heavy browsers and understates their exposure. Entering visits and its square as predictors instead lets the data determine the shape, and prior work on other online risks finds it concave (Shen and Sood, 2025).

3.6 Auditing the Audit

Two limits of the main estimates deserve scrutiny: the scan missed part of the web, and it ran long after the browsing. We examine both with two instruments independent of Blacklight, HTTP Archive and the Wayback Machine, and with two follow-up Blacklight scans: a rescan of the 500 widest-reach domains that the original pass captured, and a fresh scan of a random sample of 200 domains it missed. Overall, the estimates are lower bounds, and demographic comparisons do not depend on when tracking was measured or on how the missing scans are treated. Section C reports every check in detail.

Consider coverage first. Blacklight returned no scan for domains carrying 21% of visits, and those visits enter the main estimates as zero trackers. The domains are neither tracker-free nor dead. Parsing the scan logs shows empty returns consistent with sites blocking automated visitors (Table C.2), and Wayback Machine snapshots confirm that domains carrying most of the missed visits were live in June 2022 (Table C.3). A retry pass recovered 434 domains the first scrape had failed on, which carry 3.0% of panel visit mass and are more heavily tracked than the domains the first pass reached (11.1 against 6.9 ad trackers per domain); these are included in the estimates above. A July 2026 scan of the 200-domain sample completed for 131 of them and found at least as much tracking there as on the domains the audit reached (8.11 against 6.43 ad trackers per domain; Table C.5). For the 63% of missed visits whose domains appear in HTTP Archive's June 2022 corpus, we impute ad trackers and third-party cookies from the Blacklight counts implied by whether HTTP Archive identified trackers there (Table C.11), and we use those imputations to test the sensitivity of the demographic gaps. Coverage is also even across panelists (Section C.2) and unrelated to demographics (Table C.12), so differential missingness cannot be producing the gaps we report.

Second, the main estimates derive from a January 2025 Blacklight audit of web domains from browsing recorded in June 2022, so the audit postdates the browsing by thirty-one months. To bound the drift, we use the June 2022 and June 2025 corpora from HTTP Archive, which crawls millions of sites monthly and stores the requests each page made, matching those third-party requests to Tracker Radar to measure tracking on the panel's domains at both dates. Tracking was more prevalent in 2022 than in 2025, so a later audit understates what the panel met (Section C.3). The same two corpora let us estimate the demographic gaps at each date, and the gaps appear at both, so the demographic conclusions do not hinge on when tracking was measured (Table C.14). HTTP Archive records network requests rather than page behavior, so it is silent on session recording, keylogging, and canvas fingerprinting; for those three we rely on the April 2026 rescan of the 500 widest-reach domains, which finds domain-level prevalence close to January 2025 (Table C.8). Finally, a January 2025 HTTP Archive corpus establishes cross-instrument agreement with the Blacklight scans of the same period (Table C.9).

4 Results

4.1 Exposure to Different Kinds of Tracking

Tracking is near universal, and ad trackers and third-party cookies are the most common vehicles. Over the month, 99.1% of users encountered at least ten ad trackers or third-party cookies (Table 2). Users encountered on average 19,701 ad trackers ($\hat{\sigma} = 32{,}710$) and 23,093 third-party cookies ($\hat{\sigma} = 37{,}190$), against medians of 7,720 and 9,216, so the distributions are heavily right-skewed. Normalizing by visits removes most of the skew: users meet on average 5.2 ad trackers ($\hat{\sigma} = 3.8$) and 6.3 third-party cookies ($\hat{\sigma} = 5.2$) per visit, with medians of 4.1 and 5.0 (Table 3). Relative to their means, the rates vary about half as much as the totals, so

Table 2. Cumulative tracking exposure per panelist over the month. The final two columns give the share of panelists who encountered the technique at least once and at least ten times. Ad trackers and third-party cookies are counts of distinct third-party domains; the remaining measures are binary indicators of whether the technique was present on the domain. Visits to domains Blacklight did not scan contribute zero, so every figure is a lower bound.

Measure	Mean	SD	Min	P25	Median	P75	Max	≥ 1	≥ 10
Ad Trackers	19,701	32,710	0	2,039	7,720	22,687	326,469	99.6%	98.9%
Third-Party Cookies	23,093	37,190	0	2,589	9,216	26,186	431,493	99.4%	98.7%
Facebook Pixel	285	476	0	31	117	353	5,298	95.0%	84.9%
Google Analytics (Remarketing)	27	86	0	0	5	22	1,448	74.2%	40.6%
Session Recording	140	254	0	12	58	172	3,598	91.4%	76.9%
Keylogging	212	673	0	3	19	98	10,151	85.4%	59.8%
Canvas Fingerprinting	236	513	0	13	68	234	6,242	92.2%	78.0%

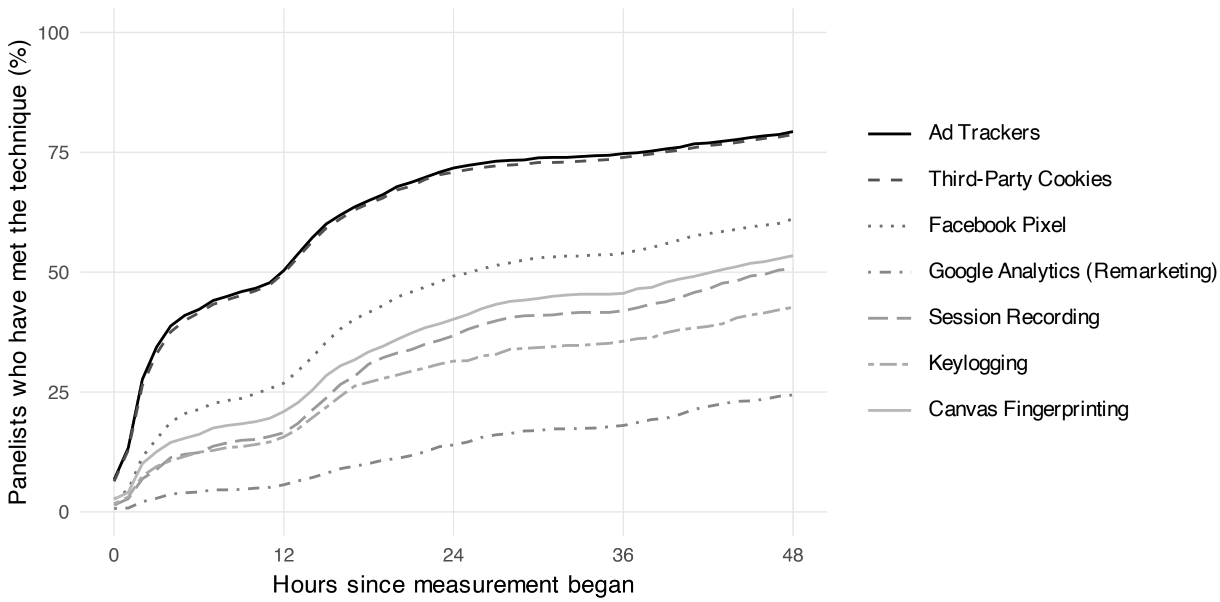
Table 3. Tracking exposure per visit. For ad trackers and third-party cookies this is the number of distinct third parties encountered per visit and can exceed one; for the binary measures it is the share of visits on which the technique was present. Ad trackers and third-party cookies are counts of distinct third-party domains; the remaining measures are binary indicators of whether the technique was present on the domain. Visits to domains Blacklight did not scan contribute zero, so every figure is a lower bound.

Measure	Mean	SD	Min	P25	Median	P75	Max
Ad Trackers	5.19	3.78	0.00	2.75	4.14	6.48	32.56
Third-Party Cookies	6.32	5.22	0.00	3.33	4.98	7.73	53.43
Facebook Pixel	0.09	0.09	0.00	0.03	0.06	0.11	1.00
Google Analytics (Remarketing)	0.01	0.04	0.00	0.00	0.00	0.01	0.99
Session Recording	0.05	0.07	0.00	0.01	0.03	0.06	0.69
Keylogging	0.04	0.08	0.00	0.00	0.01	0.04	0.60
Canvas Fingerprinting	0.07	0.09	0.00	0.01	0.04	0.08	1.00

most of the variation in how much tracking people meet comes from how much they browse rather than from where they go. Every figure here is a floor, since visits to the domains the scan did not reach contribute zero. Replacing that zero with measurement-calibrated fills raises the mean ad-tracker rate to between 6.2 and 8.1 per visit and the cookie rate to between 7.1 and 10.5 (Section C.5).

The invasive methods are met less often. Session recording, keylogging, and canvas fingerprinting are present on 4 to 9% of domains individually and on 17% taken together, and users encounter session recording on 5% of visits ($\hat{\sigma} = 7$ percentage points), keylogging on 4% ($\hat{\sigma} = 8$), and canvas fingerprinting on 7% ($\hat{\sigma} = 9$) (Table 3). Low rates still accumulate: 92.2% of users met canvas fingerprinting at least once, and 53.3% met all three at least ten times (Table 2).

Because tracking is pervasive, exposure is rapid. Using browsing timestamps, we identify when each user first encountered each tracking method. Half of the users encounter an ad tracker or a third-party cookie within the first 12 hours of the start of measurement (see Figure 1). By 48 hours, nearly 79.7% have encountered at least one tracker or cookie. Even the more intrusive techniques—session recording, keylogging, and canvas fingerprinting—reach nearly half the users within 48 hours.



(a)

	0h	12h	24h	36h	48h
Ad Trackers	0.067	0.504	0.717	0.747	0.793
Third-Party Cookies	0.064	0.498	0.708	0.739	0.787
Facebook Pixel	0.027	0.269	0.492	0.54	0.611
Google Analytics (Remarketing)	0.007	0.057	0.14	0.18	0.244
Session Recording	0.014	0.165	0.367	0.42	0.507
Keylogging	0.017	0.156	0.314	0.356	0.427
Canvas Fingerprinting	0.027	0.209	0.402	0.456	0.534

(b)

Figure 1. The proportion of users who had encountered a particular tracker by a particular time. We start measuring at 6 PM on 31 May (due to time zones) when at least 50 users have logged browsing activity. The table reports the cumulative proportions at the specified hours.

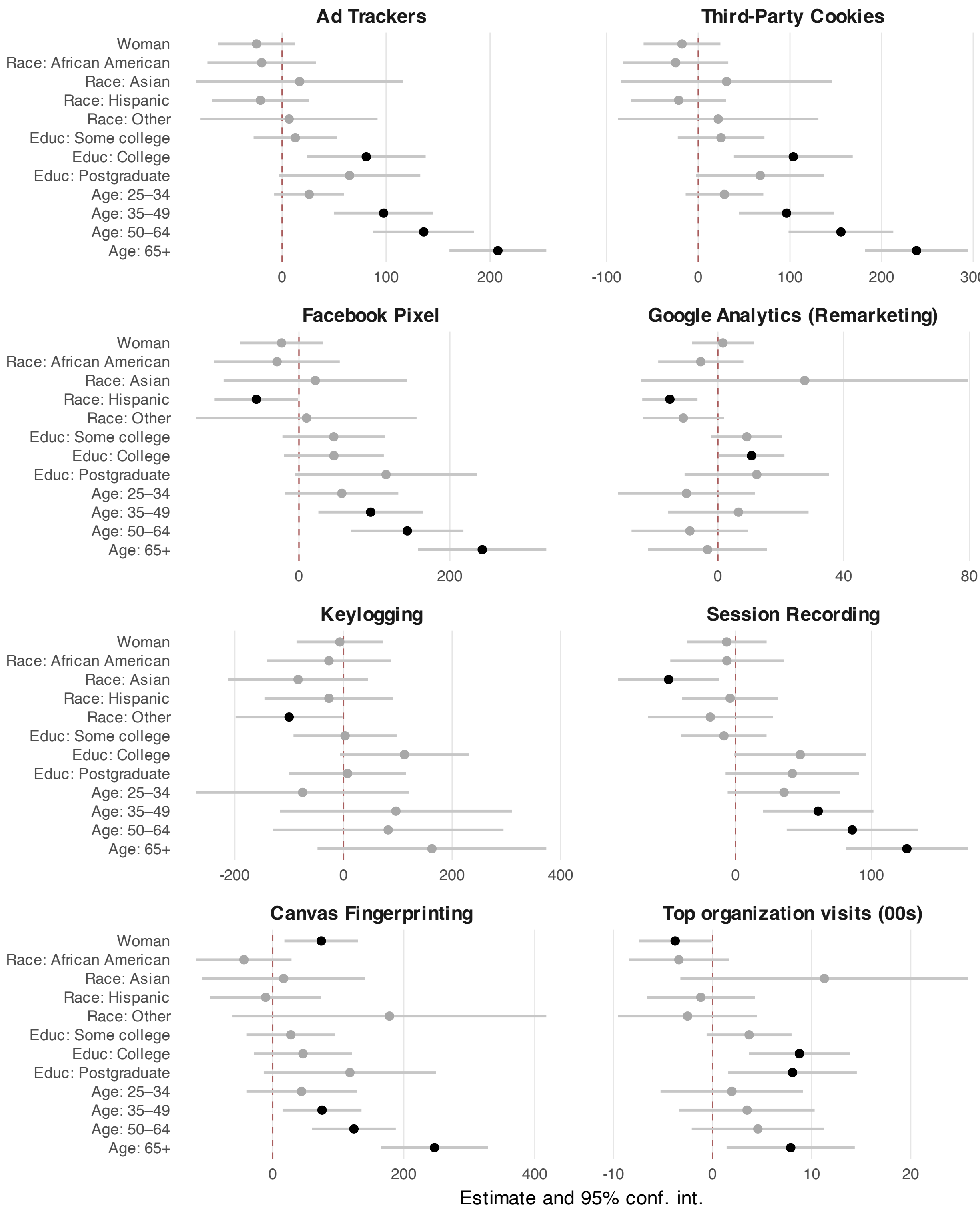


Figure 2. Demographic differences in *cumulative exposure*. Each panel reports OLS estimates (markers) and 95% confidence intervals (lines) from Equation (6) for one of the seven tracking methods, and for the number of visits observed by the top tracking organization. Black markers are significant at $p < .05$. Full tables in Section D.

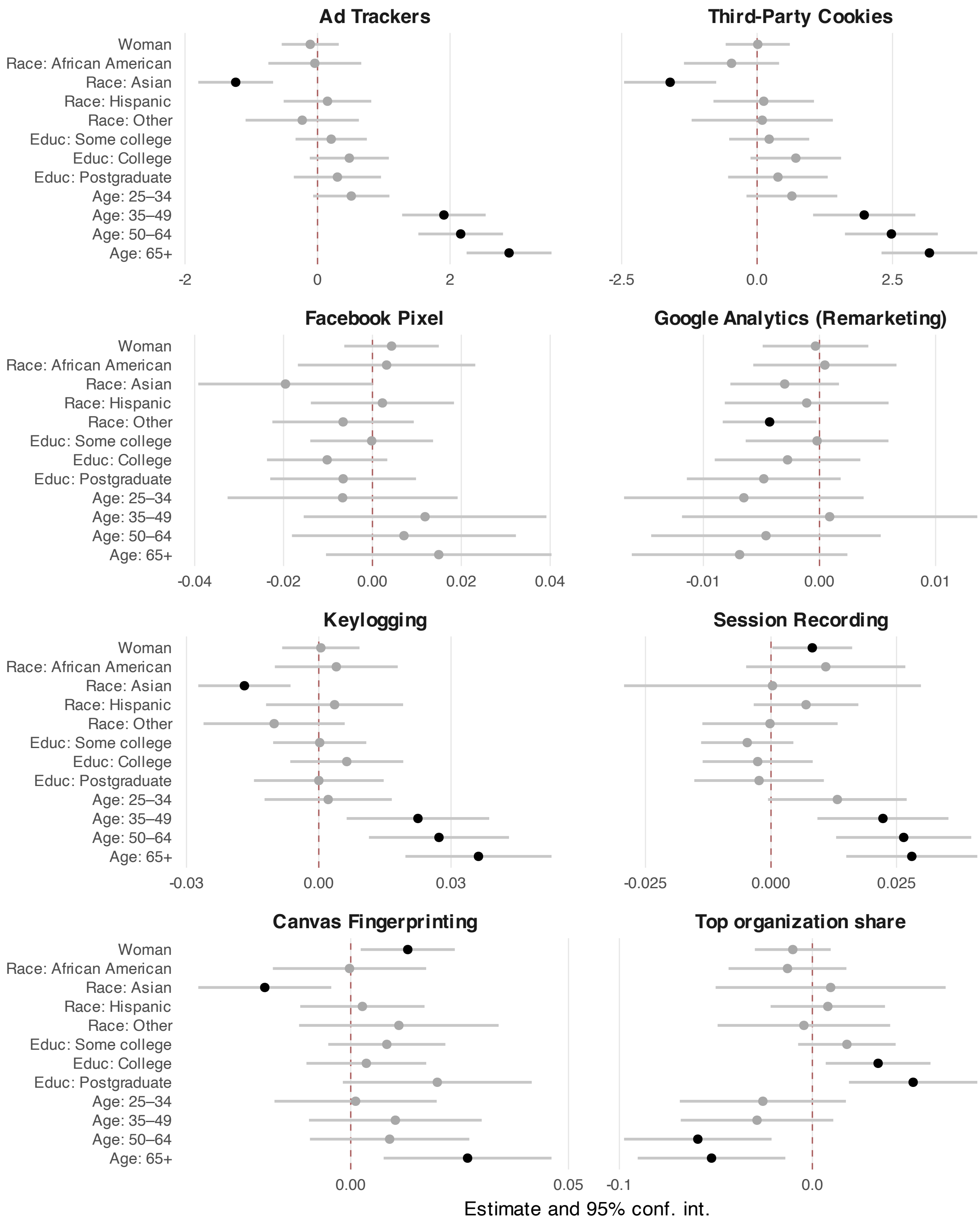


Figure 3. Demographic differences in *exposure rate*. Each panel reports OLS estimates and 95% confidence intervals from Equation (6) for the per-visit rate of each tracking method, and for the share of visits observed by the top organization. Black markers are significant at $p < .05$. Full tables in Section D.

4.2 Demographic Differences in Exposure to Tracking Methods

Before any adjustment for how much people browse, the gaps fall roughly where the opportunity account would put them (Figure 2). Gender predicts little, with two exceptions: women encounter more canvas fingerprinting than men ($\hat{\beta} = 74$, $\widehat{\mathrm{SE}} = 29$, $p < .01$), and the top organization observes fewer of their visits ($\hat{\beta} = -3.8$ hundred, $\widehat{\mathrm{SE}} = 1.9$, $p < .05$). Racial differences are confined to three cases: Asian users are less exposed to session recording, users categorized as 'Other' are less exposed to keylogging, and Hispanic users are tracked less by Facebook Pixel and Google Analytics. Education and age matter more. College-educated users encounter 10,361 more third-party cookies than users with a high school diploma or less ($p < .01$) and 48 more session recorders ($\widehat{\mathrm{SE}} = 25$, $p < .10$), with postgraduates following the same pattern. Users 65 and above encounter more trackers on every method except Google Analytics.

Expressing exposure per visit changes which of these gaps remain (Figure 3). The education gaps largely disappear: more educated users are online more, not on more heavily tracked sites. That is the pattern reported for exposure to malicious content and to breached credentials, where demographic gaps are largely a function of time spent online rather than of who is spending it.

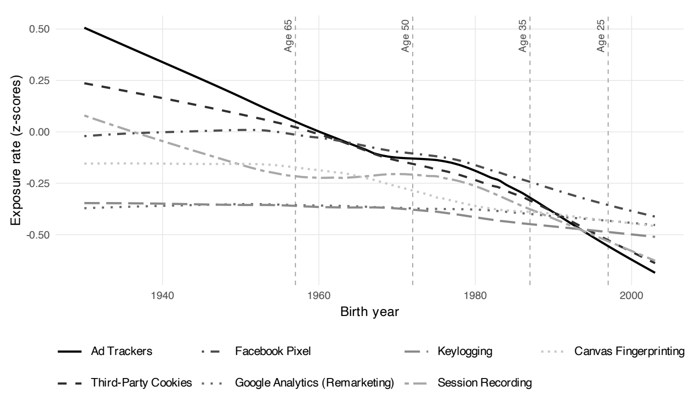


Figure 4. Exposure rate by birth year. Lines are LOWESS-smoothed standardized rates (z-scores) for the seven tracking methods, winsorized at the 95th percentile. Vertical dashed lines mark the age-group boundaries.

The age gap persists. Users 65 and above continue to meet higher per-visit rates of ad trackers, third-party cookies, session recording, keylogging, and canvas fingerprinting, and the gradient runs smoothly across birth cohorts rather than resting on where the age bins were cut (Figure 4). The gap is 79 to 91% of the under-25 mean depending on the method, a standardized effect of 0.6 to 0.8 (Section C.6). The gender gap in canvas fingerprinting also survives, at one additional fingerprinting script per 100 visits ($\widehat{\mathrm{SE}} = 0.005$, $p < .05$).[2]

Older panelists visit more heavily tracked sites, not different kinds of site. Decomposing the gap over the content categories the meter assigns, site choice accounts for 3.00 of the 3.66 ad-tracker gap and category mix for 0.26; for third-party cookies the figures are 3.20 of 4.01 and 0.30 (Section G).

The second way of holding browsing volume constant gives the same answer. Estimating Equation (7), which enters total visits and their square into a median regression on cumulative exposure, closes the education gap: the college coefficient falls from 5,559 to 68 ad-tracker encounters and is no longer distinguishable from zero ($p = .82$). Users 65 and above still meet 2,427 more ad trackers over the month than users under 25 ($p < .001$), which is 64% of the under-25 median, and 1,957 more third-party cookies ($p = .01$), or 44% (Section E). The age gap is therefore not an artifact of assuming that exposure grows in proportion to browsing.

Four features of the design could produce a gap of this kind on their own. The rate divides each panelist's encounters by their own visits, so a rate estimated from two visits carries the same weight as one estimated from forty thousand, and 90 panelists browsed fewer than a hundred times; dropping those imprecise denominators raises the 65+ ad-tracker coefficient from 2.89 to 3.02, and weighting panelists by visits, which asks about the average visit rather than the average person, raises it to 3.35 (Section F). The age bins are ours rather than the data's, so the gap could follow from where we cut them; fitting birth year as a spline leaves the gradient significant with no detectable nonlinearity, and the gap

[2] Applying a Bonferroni correction for the 12 demographic predictors tested ($p < .00417$), all coefficients with unadjusted $p < .01$ in Table D.2 remain significant.

holds within each metered device group and under a device control (Section F). The audit missed 21% of visits, and had it missed them unevenly across age groups the gap would follow from coverage rather than from exposure; per-user coverage is unrelated to every demographic we measure, and replacing the unscanned visits with calibrated fills moves the coefficient only to between 2.80 and 2.82 (Section C.6). The audit also ran thirty-one months after the browsing, so the gap might describe the web of 2025 rather than the web the panel met; measuring the same panelists against HTTP Archive at both dates leaves it significant in June 2022 as well, though roughly a quarter smaller for ad trackers and half smaller for cookies, which is consistent with tracking growing fastest on the sites older users frequent (Section C.3).

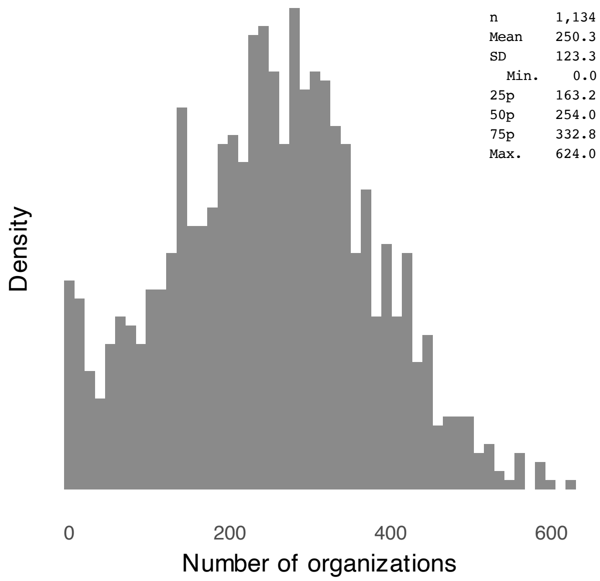
Asian users move the other way under adjustment. Lower on cumulative exposure only for session recording and keylogging, they show lower per-visit rates across nearly every web tracking method, consistent with visiting less heavily tracked sites. Under Equation (7), the Asian ad-tracker coefficient is also negative, though not distinguishable from zero (−1,024, $p = .12$).

Even so, demographics overall account for little of the total variation in tracking, under 8% across all models (Section D), so most of the inequality in surveillance lies within demographic groups rather than between them.

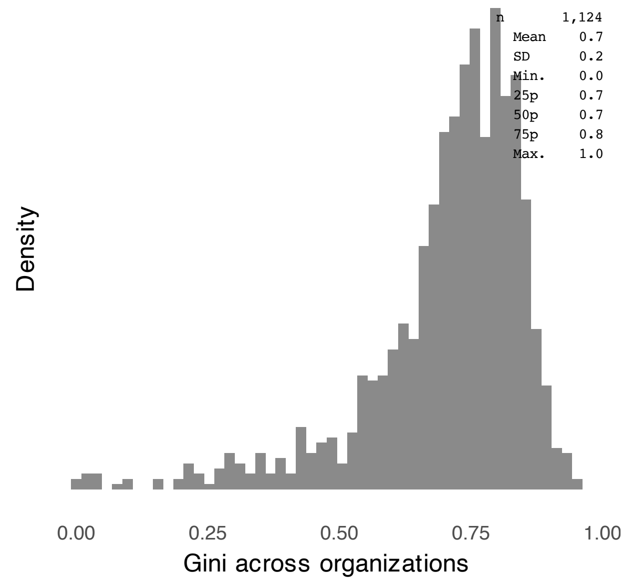
4.3 Tracking by Organizations

Trackers report to organizations, and a quantity that matters for surveillance is how much of one person's browsing a single organization can observe. Within the observation month, users are typically tracked by 163 to 333 organizations, with a median of 254 (Figure 5a). Among users tracked by at least ten organizations, a handful of organizations account for most of what is seen, with a median Gini coefficient of 0.74 (Figure 5b).

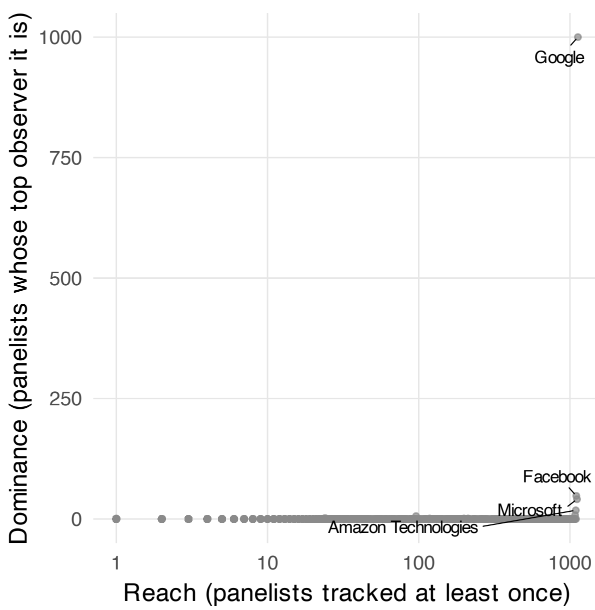
Figure 5c documents the tracking reach and dominance of organizations. Reach counts the users an organization tracks at least once. Dominance counts the users for whom it ob-



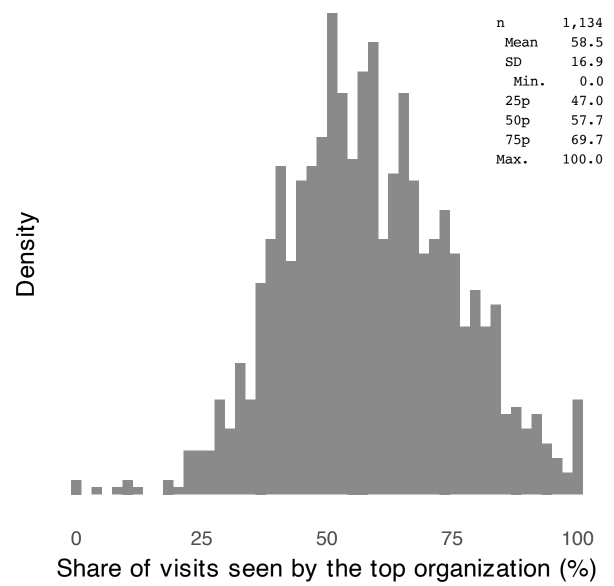
(a) Number of organizations



(b) Concentration across organizations



(c) Dominance against reach



(d) Share observed by the top organization

Figure 5. Tracking by parent organizations. Panel (a) reports how many organizations track each user's browsing. Panel (b) reports how unevenly that tracking is divided among them, as Gini coefficients, for users tracked by at least ten organizations. Panel (c) plots each organization's dominance, the number of users for whom it observes a larger share of browsing than any other organization does, against its reach, the number of users it tracks at least once; the figures in parentheses give those two counts for the labelled organizations. Panel (d) reports, for each user, the largest share of their browsing observed by any single organization.

serves more browsing than any competitor. Google leads on both, appearing in the tracker chain of 99.6% of users and standing as the largest single observer for 88.2% of them. Microsoft, Facebook, Amazon, and Cloudflare follow at a distance.[3] Google's position holds on the part of the web our scanner could not initially audit. Among a directly re-measured random sample of unscanned domains, 68 to 72% load a Google-owned third party, which is statistically indistinguishable from the 70% visit-weighted share among the domains we did scan (Section C.6).

In absolute terms, the leading organization observes a mean 56% of a user's browsing ($\hat{\sigma} = 0.17$) and a median of 54% (Figure 5d). Weighting by time spent rather than by visits gives similar figures (Section I).

4.4 Demographic Differences in Tracking by Organizations

The last question is who is most legible to the single organization that sees the most of them, which for 88.2% of users is Google. We estimate Equation (6) on the number of visits that organization observes and on the share of visits it observes (column (8) of Tables D.1 and D.2).

Women are marginally less legible than men, by 1.9 percentage points of visits observed ($\widehat{\text{SE}} = 1.0\%$, $p < .1$). College-educated and postgraduate users are more legible than users with a high school diploma or below, by 3.6 ($\widehat{\text{SE}} = 1.3\%$, $p < .01$) and 5.5 percentage points ($\widehat{\text{SE}} = 1.7\%$, $p < .01$). Both differences hold on the cumulative and the normalized measure. Racial differences are small and statistically insignificant on both measures.[4]

Age is the one place where the two measures disagree. Users 65 and above have more visits observed in total than users under 25, and a smaller share of visits observed, by at

[3] Tracking is likewise concentrated among a handful of domains (Section J). Financial and e-commerce platforms contribute heavily to session recording and keylogging exposure; Microsoft and TikTok are prominent in canvas fingerprinting.

[4] As in Section 4.2, the education and age differences persist after correcting for multiple comparisons (Footnote 2).

least 6.7 percentage points for users 50 and above ($p < .01$). We find a similar reversal under Equation (7), which holds volume constant. The total flips sign and turns marginally negative ($p < .1$; Section E). This pattern is consistent with older users browsing more and meeting more trackers per visit, but spreading their browsing across sites in a way that leaves less of it under any single observer.

As with exposure (Section 4.2), most of the variation in who is watched, like most of the variation in how much, with demographics explaining under 4% of the total variation (Tables D.1 and D.2).

5 Discussion

Two accounts of who bears the most risk online have been tested against observed behavior, and for malicious content and for breached credentials, the second has prevailed: exposure follows opportunity, and demographic gaps close once time spent online is taken into account. Web tracking is the case where that account runs out. Adjusting for browsing volume closes the education gap here as it does elsewhere, but leaves the age gap standing. Users 65 and older meet 76 to 87% more trackers per visit than users under 25, a standardized gap of 0.6 to 0.8.

The reason is structural rather than statistical. Malicious content can be recognized and avoided, and people do avoid it. Breached credentials follow from accounts a person opened. Tracking is installed by the sites a person visits, under arrangements between those sites and third parties the visitor never chose, and it runs whether or not the visitor knows it is there. Skill and caution have purchase on the first two risks and almost none on the third, so the composition of a person's browsing survives as a source of inequality where for other risks it does not. That older users are the group left exposed is the part with practical weight: the gap runs across five of the seven techniques we measure, including session recording, keylogging, and canvas fingerprinting, which the hygiene measures ordinarily recommended to users do not reach.

The same panel carries a security measure for the same people over the same month, and it points the other way. Antivirus vendors flagged some of the domains these panelists visited as malicious or suspicious, and exposure to those domains falls as age rises exactly where tracking exposure climbs (Table 4). Users under 25 meet the fewest ad trackers per visit and the largest share of flagged domains; users 65 and above meet the most trackers and the smallest share. Across the five age groups the two gradients are almost perfectly opposed (Spearman -0.90, $p = .08$ on five points). We report this as a fact about composition, not about individuals: at the person level, once browsing volume and demographics are held fixed, the association between the two risks is not robust to how each is normalized, and we do not claim one.

That the two risks land on opposite groups is what the structural account predicts. Avoidance is available against malicious content and is practised more by the users who have more of the skills that avoidance requires; it is not available against tracking, so tracking accumulates wherever browsing goes.

Table 4. Two online risks, opposite age gradients

Age group	n	Ad trackers per visit	Flagged domains (%)
<25	106	3.32	12.3
25–34	200	3.94	10.8
35–49	291	5.32	8.6
50–64	280	5.61	9.0
65+	257	6.31	8.3

Note: Same 1,134 panelists and the same observation month for both measures. "Flagged domains" is the share of a panelist's distinct visited domains that antivirus vendors classified as malicious or suspicious (Shen and Sood, 2025). Spearman correlation across the five group means is -0.90 ($p = .08$); with five points it describes the ordering rather than testing it.

The magnitude deserves a benchmark, because effect sizes in this literature rarely have one. The gender gap in per-visit exposure is small and, apart from canvas fingerprinting and session recording, statistically indistinguishable from zero, which puts it an order of magnitude below the ten to twenty percent gender gap in earnings documented for highincome countries (Goldin, 2014). The age gap is several times larger in relative terms than that earnings gap. And as most of the residual gender pay gap sits within occupations rather than between them, most variation in surveillance sits within demographic groups rather than between them: our covariates account for under 8% of the variance. Who a person is explains little about how much surveillance they meet. What they browse explains a great deal.

Two further results bear on how the tracking ecosystem is organized. Exposure is near universal and it arrives fast. More than 99% of users met ad trackers and third-party cookies, half of them within twelve hours of the start of measurement. And exposure is concentrated: users are tracked by a median of 254 organizations, but a single organization observes a median 58% of a person's browsing, and for 88.2% of users that organization is Google. Microsoft and Facebook are dominant for roughly four percent of users each. We recover the same leading organizations as Dambra et al. (2022), who aggregate over visits rather than over users, and the user-level measure adds what aggregation cannot: the concentration is not an artifact of a few heavy browsers but the ordinary condition of the median person. It also extends to the part of the web our scanner could not reach, where 68 to 72% of a directly re-audited sample of domains load a Google-owned third party.

Every estimate here is a floor, and deliberately so. Visits to domains the audit could not complete contribute nothing, and a hand-coded random sample of those domains, rescanned directly, carries more tracking than the domains we did scan. Replacing the zero with measurement-calibrated imputation puts the mean ad-tracker rate between 6.2 and 8.0 per visit against the 5.2 we report. The demographic comparisons hold under every fill we can defend.

Tracking data enables stalking, identity theft, and reputational injury (Citron and Solove, 2022). It steers people in financial hardship toward high-interest credit (Christl and Spiekermann, 2016); it has been used to withhold housing and employment advertising by race and location (Angwin and Parris, 2016); and a handful of behavioral traces suffices to re-identify a person in data stripped of identifiers (Achara et al., 2015; Su et al., 2017).

Users can defend against only part of such tracking. Cookies can be blocked or deleted, and the major browsers have moved against third-party cookies. On the other hand, the invasive techniques leave nothing to block or delete. Session recording scripts capture mouse movement and form input as it happens (Senol et al., 2022), keyloggers take keystrokes before a form is ever submitted, and canvas fingerprinting identifies a device while storing nothing on it (Acar et al., 2014; Mowery and Shacham, 2012).

What follows for practice runs in three directions. Protection is worst matched to need: the heaviest per-visit exposure falls on the users least likely to seek out defenses, and it falls in part through techniques that adopting defenses would not stop, which is an argument for protections that ship enabled by default in the browser rather than for tools users must find and install. Regulators concerned with a single firm's visibility into a population's browsing now have a measured baseline rather than an inference from crawls, and it is a lower bound. And the drift we document, an age skew that widened between June 2022 and June 2025 on an instrument held constant, is a quantity no site-centric audit can produce, which argues for treating exposure measurement as a recurring exercise rather than a one-off.

5.1 Limitations

The audit is browser-based, so it sees nothing of the tracking inside mobile applications, which runs through embedded SDKs (Achara et al., 2015; Binns et al., 2018). Restricting the analysis to the 673 panelists whose logged activity is desktop only, where metering is most complete, leaves the demographic patterns in place (Section C.7), but that check conditions on the metered device and cannot speak to application traffic.

Blacklight has blind spots of its own. It does not detect CNAME cloaking, and it does not see server-side collection that occurs outside the browser even when a user blocks cookies. It also does not distinguish tracking from adjacent uses of the same technology: session recording and canvas fingerprinting are sometimes deployed for bot detection or interface testing rather than surveillance (Mattu and Sankin, 2020; Senol et al., 2022). Our estimand is defined accordingly, as the client-side tracking a user's own browser could in principle observe.

The audit-based measures are presence-weighted. They record the tracking technology present on a visited domain, not a confirmed capture. For always-on technologies the two coincide. For interaction-dependent technologies such as keylogging, presence bounds collection from above while the zero-filled visits bound presence from below. Presence is the quantity a site controls and a privacy audit can verify, and it is the proxy the literature has used (Dambra et al., 2022; Karaj et al., 2019; Mattu and Sankin, 2020; Niforatos et al., 2021; Zheutlin et al., 2022b,a).

The sample is matched to US population margins but not identical to them. Gender, education, and age margins are statistically indistinguishable from the Current Population Survey; race is not, with a mean absolute deviation of 1.2 percentage points across sixteen margins (Table 1).

Finally, passive metering is imperfect in two further respects. Panelists browse on devices and in browsers the meter does not cover (Bosch et al., 2025), and knowing they are metered may suppress the browsing they do. Both push measured exposure down.

6 Conclusion

Nearly everyone is surveilled on the web, most of them within a day of being observed, and for nine users in ten a single company sees more than half of where they go. Those facts were widely suspected and are now measured on a sample built to stand for a population. The finding that does not follow from what came before concerns who bears the burden. For other online risks, demographic gaps in exposure are largely a matter of how much time people spend online. For surveillance they are not: the gap by age survives adjustment for browsing volume, runs across every technique we measure, and widened over the three years our instruments span. Exposure to surveillance is not something users select into, and it is not distributed the way the risks they can select into are. The browsing panel, the audits, and the analysis code are released so that the estimate can be improved on, and repeated.

Ethics

The study protocol was reviewed by the Colorado State University Institutional Review Board. The browsing data used in this study were passively collected, fully anonymized, and contained no personally identifiable information. No IRB approval was required.

Code and Data

Code and data is available under an open source license at https://github.com/themains/private_blacklight.

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Supporting Information

A Participant consent and data privacy

Before enrolling in a YouGov panel, people receive detailed information about the nature and scope of the data collection. Potential participants are informed about the types of data that will be collected, such as visited domains, and what will not be collected, including any information entered into secure forms, such as usernames, passwords, or payment details (See YouGov's FAQ, https://today.yougov.com/about/faq).

Only after reviewing this information do individuals consent to participate. Participation is entirely voluntary, and panelists can pause or uninstall the tracking software at any time. (See pages 3 and 4 of the installation guide for Terms and Conditions and Privacy Policy made known to participants.)

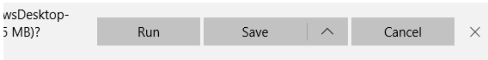
The browsing data collection application—YouGov Pulse—is developed in partnership with RealityMine and is available as a browser extension or mobile app. The app ensures anonymity: researchers never have access to identifying information, and no data is shared with third parties.

To encourage participation, panelists earn points through YouGov's reward system—2,000 points upon joining and an additional 1,000 points for completing a full month of activity.

Installing YouGov Pulse on Windows

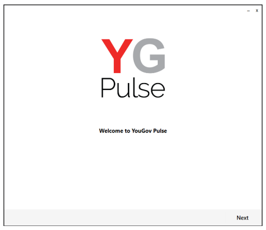
Step 1

Open the link provided from either the survey or the email. Download the software and click "Run" (depending on your browser, this might look slightly different) or open the file.



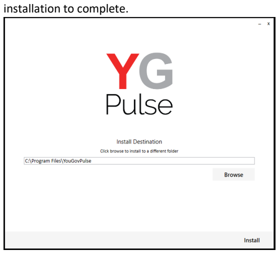
Step 2

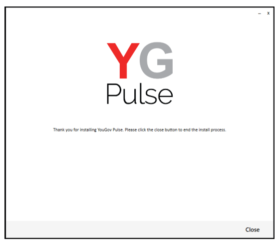
Start the installation process and click "Next"



Step 3

Choose installation destination and click "Install". Accept any prompts from Windows to allow installation to complete.





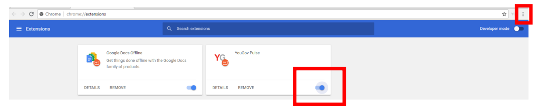
Step 4

Install the Google Chrome and/or Mozilla Firefox browser extension(s) by clicking 'OK' on the box that pops up – if either of them are open (Note: The browser will close.)

If the browsers are not open, you will not see the message box. Instead, you will notice that the extension has been added next time you open the browser.

On Chrome-

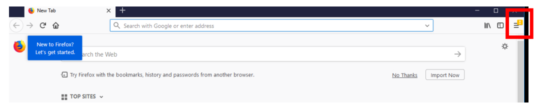
Either after you have clicked 'OK' or next time you open Google Chrome, click "Enable Exernsion" on the window in the top right. If you can't see this notification, click on the three dots next to the URL bar and select "More Tools > Extenstions". Ensure that the YouGovPulse Extension is enabled by moving the slider to the right if necessary:



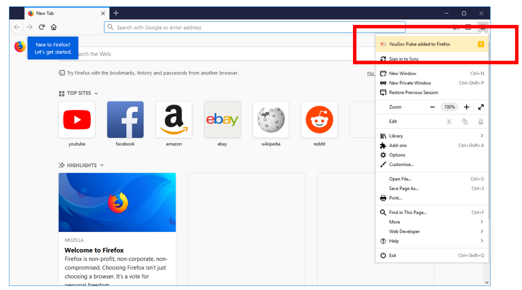
Once it is installed and enabled, you will see the YouGov Pulse icon to the right of the URL bar.

On Firefox-

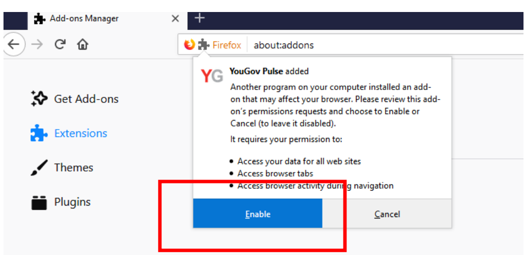
Either after you clicked "OK" or next time you open Mozilla Firefox, click the yellow exclamation mark under the "Open Menu" icon:



Click on this, then select "YouGov Pulse added to Firefox"

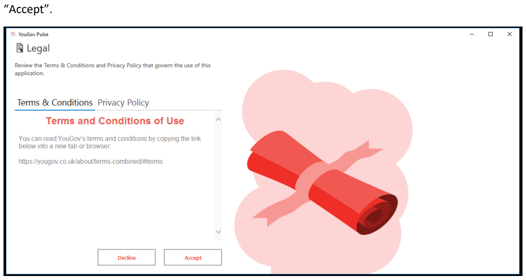


and "Enable" to activate the Add-On.



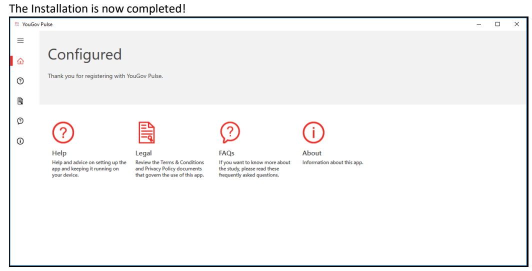
Step 5

Open the YouGov Pulse App, read the Terms and Condition and the Privacy Policy and select "Accept".



Completed!

The Installation is now completed!



IMPORTANT NOTE: *Please make sure that the software is running in the background at all times. If the app stops running, you'll stop earning your points. You can delete the App at any time if you decide to no longer be part of the YouGov Pulse project.*

B Tracking methods

This appendix summarizes the seven tracking methods that Blacklight detects on the homepage of the domain and one additional randomly selected internal page (Mattu and Sankin, 2020). Blacklight analyses are retrieved from a 24–48-hour cache when available or performed in real-time if no recent results exist. Scans run in a headless Chromium 126 browser emulating an iPhone 13 Mini user agent, from an AWS US-California environment, with a 30-second per-page timeout, a network-idle wait condition, a fresh browser cache per scan, and with HTTP Archive capture enabled. The emulated device matters little for what a scan detects. HTTP Archive crawled these domains in June 2022 as both a phone and a desktop browser, finding ad trackers on 94.0% of them on desktop and 94.1% on mobile (Table C.6). The main scans were done in January 2025. We assess the temporal stability of the Blacklight analyses using a new audit in April 2026 for 500 of the top domains in terms of reach, covering 99.6% of participants, 58.5% of visits, and 58.5% of browsing time. With the exception of Facebook Pixel, we find similarly high (or higher) prevalence rates across categories (Jan 2025 vs. Apr 2026): Ad Trackers (67.4% vs. 67.2%), Third-party Cookies (57.6% vs. 53.6%), Facebook Pixel (21.6% vs. 16.6%), Google Analytics (2.8% vs. 23.8%), Session Recording (10.0% vs. 9.2%), Keylogging (3.8% vs. 4.2%), and Canvas Fingerprinting (12.4% vs. 20.6%).

- **Ad Tracking:** Ad trackers are third-party scripts embedded in websites that collect user browsing behavior and send it to advertising networks. These scripts help build user profiles for targeted advertising or retargeting across websites. Blacklight detects ad tracking by identifying network requests to known advertising domains (domains under "Ad Motivated Tracking", https://github.com/duckduckgo/tracker-radar/blob/main/docs/CATEGORIES.md) in the DuckDuckGo Tracker Radar list.

- **Third-party Cookies:** Cookies are small text files stored in the user's browser. Third-party cookies originate from domains other than the one being visited and are widely used to track users across websites.

- **Facebook Pixel:** Facebook Pixel is a tracking script that monitors user behavior—such as page views, button clicks, and purchases—and sends this data to Facebook for ad targeting and conversion analytics. It links off-site behavior to user profiles across the Facebook ecosystem, even if users are not logged in to Facebook. Blacklight detects Facebook Pixel by identifying network requests to Facebook domains and inspecting URL query parameters for data patterns that match Pixel's documented schema.

- **Google Analytics (Remarketing):** Google Analytics uses JavaScript tags and cookies to monitor user behavior such as session duration, navigation, and referrals. Blacklight does not flag Google Analytics itself; it flags the optional "remarketing audiences" feature, which turns that measurement into cross-site advertising targeting. Detection is a request to `stats.g.doubleclick.net` carrying a Google account identifier (a `UA-`, `G-` or `AW-` prefix). A positive therefore implies both Google Analytics and remarketing; a negative means remarketing was not detected, not that Google Analytics is absent.

- **Session Recording:** Session replay scripts record user activity on a website, including mouse movements, scrolling, and form inputs—often in real time (Senol et al., 2022). These recordings can be replayed by website owners, revealing detailed behavioral data and potentially sensitive information. Blacklight detects session recording by monitoring network requests for URL substrings known to be associated with session replay tools (`https://web.archive.org/web/20210830151649/https://gist.github.com/gunesacar/0c67b94ad415841cf3be6761714147ca`).

- **Keylogging:** A potentially more invasive subset of session recording, keylogging captures every keystroke a user makes—including input into masked fields like passwords and credit card forms—before submission. This technique can reveal highly sensitive user data. Blacklight enters pre-determined text into input fields and monitors network requests for the same outgoing data.

- **Canvas Fingerprinting:** This method leverages the HTML5 canvas element to render invisible graphics and analyze subtle rendering differences based on the user's hardware and software configuration (Acar et al., 2014; Mowery and Shacham, 2012). These differences can be used to create a persistent, stateless identifier for tracking users across sessions (Karaj et al., 2019; Mattu and Sankin, 2020; Pugliese et al., 2020). Blacklight infers that canvas fingerprinting is used for tracking if scripts silently draw meaningful content on a sufficiently large canvas, do not use it for interactivity, and then extract pixel-level data in a way consistent with generating unique user identifiers.

C Measurement Validity

Two features of our design warrant scrutiny: browsing was observed in June 2022 but the Blacklight scans ran in January 2025, and the scans completed for 53.9% of domains covering 79.0% of visits. We examined both with independent instruments—HTTP Archive's monthly crawls, Wayback Machine snapshots from the browsing window—and a fresh Blacklight scan of a random sample of unscanned domains—asking not whether the paper's numbers survive but where they move and by how much. Section C.1 characterizes the unscanned domains. Section C.2 shows that coverage is similar across users. Section C.3 bounds temporal drift. Section C.4 validates the auxiliary instruments where we rely on them. Section C.5 replaces the zero-fill for unscanned visits with calibrated imputation. Section C.6 tests whether either compromise moves the demographic comparisons. Section C.7 restricts to desktop-only panelists.

Throughout, one asymmetry should be kept in view. The behavioral detections—canvas fingerprinting, session recording, and keylogging—require executing a live page and cannot be observed in request maps or static snapshots, so the auxiliary instruments are silent on them. For these three, the zero-fill's conservatism and the internal rescan in Section C.3 are the evidence.

C.1 Scan Coverage and Selection

Blacklight returned usable results for 34,512 of the 64,074 unique domains. If the failures were concentrated among heavily visited sites, the exposure measures would inherit a bias no reweighting could fix. They are not. Table C.1 reports the share of domains scanned at increasing thresholds of reach, the number of panelists who visited the domain at least once. The share rises with reach, from 53.9% across the full set to 57.2% among domains visited by at least ten panelists and 79.2% among the 130 visited by a hundred or more. Popular sites are, if anything, easier to scan. Weighted by traffic, the scanned set covers 3,477,143 of 4,398,822 visits, or 79.0%.

Why did the other half fail? Parsing the scraper's error log attributes nearly all failures to the domain side rather than to our infrastructure (Table C.2): 95.2% of unscanned domains, carrying 78.3% of unscanned visits, returned an empty scan result consistent with bot walls, JavaScript challenges, or non-HTML endpoints; 4.6% reflect scraper-to-API network failures; and 49 domains were unreachable at scan time, though these include large live sites and carry 13.4% of unscanned visits.

Nor were the unscanned domains dead during the browsing window. Wayback Machine CDX queries establish that domains carrying at least 65.8% of the unscanned visit

Table C.1. Blacklight scan success by domain reach

Reach threshold	Domains	Scanned
$\geq$ 1 panelist	64,074	53.9%
$\geq$ 2 panelists	17,828	54.6%
$\geq$ 5 panelists	5,800	55.5%
$\geq$ 10 panelists	2,745	57.2%
$\geq$ 50 panelists	387	64.1%
$\geq$ 100 panelists	130	79.2%

Note: Reach is the number of panelists who visited the domain at least once during June 2022.

Table C.2. Why scans failed

Failure reason	Domains	Visits	% of unscanned domains	% of unscanned visits	% of all visits
Scan returned nothing (bot wall / JS challenge / non-HTML)	28,149	722,044	95.2	78.3	16.4
Scraper-to-API network failure (not a domain property)	1,364	76,308	4.6	8.3	1.7
Site unreachable at scan time	49	123,327	0.2	13.4	2.8

Note: Failure causes parsed from the January 2025 scraper error log for the 29,562 domains without a completed scan.

mass were demonstrably alive in mid-2022; among unscanned domains with any 2022 check, the alive share is 80.6% (Table C.3).

Table C.3. Liveness of unscanned domains in mid-2022

Set	Domains	% of all visits	% domains alive	% visit mass alive
all BL-unscanned	29,562	21.0	22.1	65.8
BL-unscanned with a 2022 check (HA or WB queried)	8,099	16.8	80.6	82.2

Note: Liveness established via Wayback Machine CDX snapshots dated mid-2022. Shares are lower bounds; a domain without a snapshot is not necessarily dead.

Finally, we audited a seeded random sample of 200 unscanned domains directly: 100 drawn proportional to visits and 100 uniformly, each hand-coded against a written rubric (`selection_audit/CODING_RUBRIC.md` in the replication materials), probed, and freshly scanned with Blacklight in July 2026. By visit mass, the unscanned web is mostly ordinary user-facing content (61%, CI $[51, 70]$) plus ad-tech and CDN infrastructure (28%); dead domains account for 7% (Table C.4). Of the sample, 66% produced completed scans in 2026, and tracking on those domains is heavier than in the scanned population: 8.11 ad trackers per visit-weighted domain (CI $[4.61, 12.14]$) against 6.43, and 9.49 third-party cookies (CI $[4.29, 16.12]$) against 7.53 (Table C.5, Figure C.1). Compared against the calibrated fills of Section C.5 on the 43 sampled domains that procedure actually fills, direct measurement runs higher, not lower: the fill predicts 6.81 ad trackers where fresh scans find 11.05. The out-of-sample check therefore says the calibration is conservative, which is the same direction as every other strand here.

Table C.4. Composition of the unscanned-domain audit sample

	Visits stratum			Uniform stratum		
Category	n	%	95% CI	n	%	95% CI
content_site	61	61	[51, 70]	71	71	[61, 79]
adtech_infrastructure	20	20	[13, 29]	3	3	[1, 8]
cdn_api_host	8	8	[4, 15]	3	3	[1, 8]
dead	7	7	[3, 14]	15	15	[9, 23]
unknown	3	3	[1, 8]	6	6	[3, 12]
adult_content	1	1	[0, 5]	1	1	[0, 5]
parked_or_forsale	0	0	[0, 4]	1	1	[0, 5]

Note: 2×100 unscanned domains, hand-coded against a written rubric. The visits stratum is drawn proportional to visit mass; the uniform stratum is drawn uniformly over unscanned domains.

Table C.5. Tracking on freshly scanned unscanned domains

	Scanned pop.	Audit sample (direct scan)		Strand-B fill vs. direct			Uniform
Measure	(visit-wtd)	Mean	95% CI	Predicted	Direct	n	mean
Ad trackers	6.43	8.11	[4.61, 12.14]	6.81	11.05	43	8.04
Third-party cookies	7.53	9.49	[4.29, 16.12]	6.71	14.78	33	10.82
Facebook Pixel	0.09	0.09	[0.03, 0.16]	0.09	0.1	43	0.22
Google Analytics (Remarketing)	0.01	0.27	[0.16, 0.38]	0.01	0.33	43	0.38

Note: Means per domain. Audit domains scanned with Blacklight in July 2026; the scanned population reflects January 2025 scans. The audit mean is a Hájek estimator using inclusion probabilities simulated under the sampling design that drew the sample. "Predicted" is the calibrated fill of Section C.5, and "Direct" the fresh scans, both restricted to the n sampled domains that HTTP Archive measured in June 2022 — the only domains for which that fill is defined.

C.2 Per-User Coverage

Coverage of 79.0% of visits is a pooled figure, and a pooled figure is consistent with some panelists being covered almost completely and others hardly at all. Because every quantity in the paper is measured at the user level, that heterogeneity would matter. It is small. Across the 1,134 panelists, the share of visits landing on a scanned domain has a mean of 77.4%, a median of 80.2%, and a standard deviation of 14.7 percentage points; the fifth and ninety-fifth percentiles are 49.3% and 96.1%. Put differently, 94.8% of panelists have at least half their visits covered and 75.0% have at least 70% covered. Section C.6 shows coverage is also unrelated to demographics.

C.3 Temporal Drift

The browsing predates the scans by thirty-one months. We bound the drift over that interval in two ways: with a single independent instrument measured at both dates, and with an

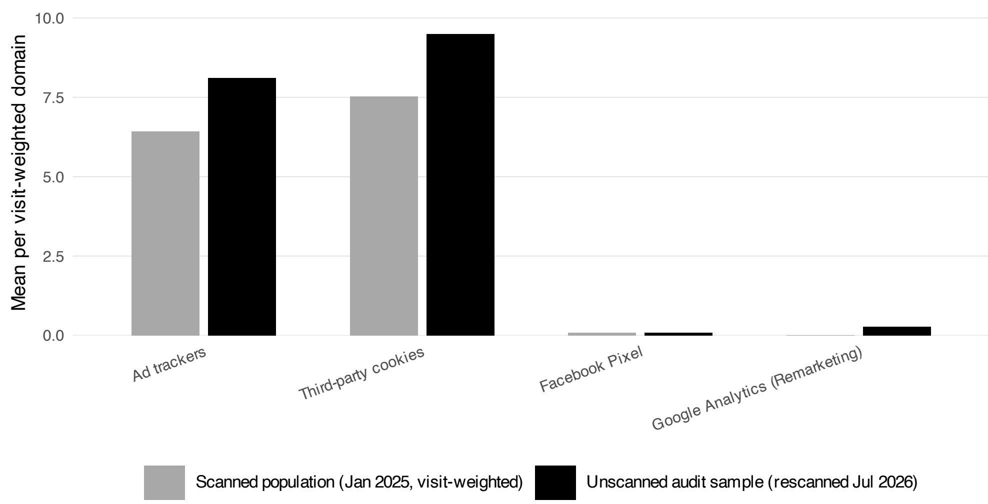


Figure C.1. Tracking on the audit sample of unscanned domains: direct 2026 scans versus the scanned population and the calibrated fill.

internal Blacklight rescan.

Same-instrument drift. HTTP Archive crawls millions of origins monthly and stores every request. For the 11,965 panel domains (desktop; 11,944 mobile) present in both the June 2022 and June 2025 crawls, we classify third-party requests against tracker lists and compare (Table C.6). Visit-weighted ad-tracker prevalence declined from 97.3% to 94.2%; the share of domains with any known tracker fell 0.8 percentage points unweighted and 2.1 visit-weighted. Header-set third-party cookies fell 3.1 points unweighted and 4.2 visit-weighted, and Facebook requests considerably more, consistent with the industry's retreat from third-party cookies and tighter pixel governance. The cookie comparison runs on the 3,843 desktop domains (3,891 mobile) for which the crawl's cookie extract queried both dates; restricting to those is necessary because coverage of that extract is itself not stable across crawls, and treating an unqueried domain as one with no cookies would manufacture a decline out of the coverage change alone. In every category, June 2022 tracking was at least as prevalent as later measurement suggests: the delayed scans understate exposure during the browsing window.

Drift at the domain level matters only insofar as it moves user-level quantities. Recomputing each panelist's exposure under the June 2022 versus June 2025 domain measurements shifts means modestly for three of the four measures—ad trackers −7.1%, Google Analytics −4.8%, Facebook requests −30.0%—while header-set cookies rise +26.1%, and the cross-user ordering the demographic analyses rely on is preserved throughout, with correlations of 0.72 to 0.82 (Table C.7, Figure C.2).

Table C.6. Same-instrument tracking drift on panel domains, June 2022 to June 2025 (HTTP Archive)

			Share of domains (%)			Visit-weighted (%)		
Platform	Measure	N	2022	2025	Δ	2022	2025	Δ
desktop	Ad trackers	11,965	94.0	91.8	-2.2	97.3	94.2	-3.1
desktop	Third-party cookies (header-set)	3,843	87.4	84.3	-3.1	96.4	92.2	-4.2
desktop	Facebook requests	11,965	56.0	45.8	-10.2	70.2	40.3	-29.9
desktop	Google Analytics/GTM	11,965	89.3	85.8	-3.5	88.0	84.3	-3.7
desktop	Any known tracker	11,965	97.9	97.1	-0.8	99.2	97.1	-2.1
desktop	Any third party	11,965	98.2	97.6	-0.6	99.4	99.0	-0.4
mobile	Ad trackers	11,944	94.1	91.9	-2.2	97.2	94.4	-2.8
mobile	Third-party cookies (header-set)	3,891	86.6	83.5	-3.1	95.9	91.6	-4.3
mobile	Facebook requests	11,944	55.5	45.5	-10.0	49.6	40.8	-8.7
mobile	Google Analytics/GTM	11,944	89.3	85.7	-3.6	87.9	84.1	-3.9
mobile	Any known tracker	11,944	97.9	97.1	-0.8	99.1	97.1	-2.1
mobile	Any third party	11,944	98.2	97.6	-0.7	99.3	98.9	-0.4

Note: Panel domains present in both crawls. Constructs are HTTP Archive's: header-set cookies are a subset of all third-party cookies; Google Analytics/GTM is presence, not remarketing.

Table C.7. User-level exposure under June 2022 versus June 2025 measurement (HTTP Archive)

Measure	Mean, 2022	Mean, 2025	Δ (%)	r
Ad trackers	22.806	21.196	-7.1	0.82
Third-party cookies (header-set)	14.976	18.887	+26.1	0.75
Facebook requests	0.596	0.417	-30.0	0.72
Google Analytics/GTM	0.882	0.840	-4.8	0.78

Note: Per-user visit-weighted rates computed with the same browsing data and the same instrument, swapping only the measurement date. r is the cross-user Pearson correlation between the two.

Internal rescan. We rescanned with Blacklight in April 2026 the 500 widest-reach domains among those scanned successfully in January 2025; the 500 cover 99.6% of panelists, 58.5% of visits, and 54% of the mean panelist's visits. Net prevalence is stable for the categories carrying most of our estimates: ad trackers move 0.2 percentage points, session recording 0.8, keylogging 0.4; third-party cookies fall 4.0 points and Facebook Pixel 5.0 (Table C.8). Canvas fingerprinting rises 8.2 points and Google Analytics remarketing 21.0. We do not read these last two as web drift. The 2026 audit ran Blacklight 3.10.0 under Chromium 138 against 3.4.0 under Chromium 126 in 2025, and detection for both categories changed in between; Google Analytics is the one category where we decline to interpret the change at all. Two categories absent in 2025 appear in 2026, TikTok Pixel on 3.8% of domains and X Pixel on 6.6%, a reminder that the tracker inventory is itself not fixed. This rescan is also the only temporal evidence available for the behavioral detections, and it shows their

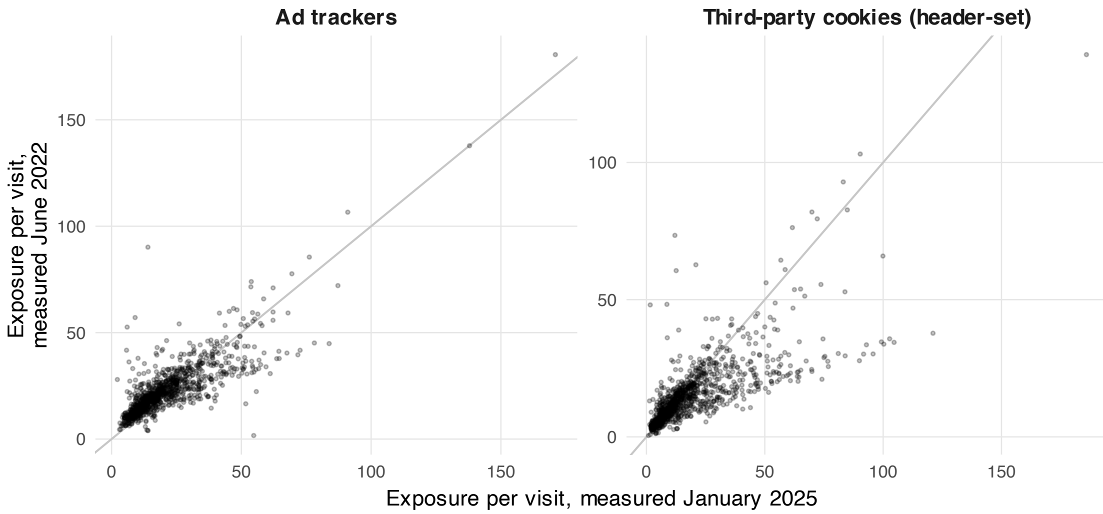


Figure C.2. Per-user exposure under June 2022 versus June 2025 domain measurement.

domain-level flags stable in aggregate.

Aggregate stability conceals domain-level churn. Agreement between the two scans runs from 75.2% for third-party cookies to 94.4% for keylogging, but persistence among January positives is lower for rare categories: 56.0% for session recording and 31.6% for keylogging (Table C.8). Misclassification of this kind is close to classical; it attenuates the demographic coefficients rather than manufacturing them, and exposure aggregates over thousands of domains per user.

Table C.8. Blacklight rescan of the top-500 panel domains, January 2025 versus April 2026

	Jan 2025 (%)	Apr 2026 (%)	Change (pp)	Agreement (%)	Persistence (%)
Ad trackers	67.4	67.2	−0.2	83.4	87.5
Third-party cookies	57.6	53.6	−4.0	75.2	75.0
Facebook Pixel	21.6	16.6	−5.0	83.8	50.9
Session recording	10.0	9.2	−0.8	92.0	56.0
Key logging	3.8	4.2	+0.4	94.4	31.6
Canvas fingerprinting	12.4	20.6	+8.2	85.8	75.8
Google Analytics	2.8	23.8	+21.0	76.2	50.0

Note: $N = 500$. Agreement is the share of domains with the same label in both scans; persistence is the share of January positives still positive in April. The 2026 audit used Blacklight 3.10.0 on Chromium 138 against 3.4.0 on Chromium 126 in 2025; the canvas fingerprinting and Google Analytics changes coincide with the instrument change and should not be read as web drift.

C.4 Instrument Agreement

Strands using HTTP Archive and Wayback are only as good as those instruments' agreement with Blacklight where they measure the same thing at the same time. On the construct they share, they agree: for ad-tracker presence, HTTP Archive and Blacklight coincide on 85.6% of the 7,420 jointly measured domains (January 2025, mobile emulation on both sides), with Spearman $\rho = 0.61$ on counts (Table C.9). Where the constructs differ, the labels say so: header-set cookies are a subset of all third-party cookies, and Google Analytics/GTM presence is a much broader construct than Blacklight's remarketing detection. On the 2,602 domains where the crawl actually queried cookies, the two instruments still agree on presence 74.2% of the time ($\rho = 0.52$); the Google Analytics pair agrees on 21.0% ($\rho = 0.07$), and it is that pair we decline to use as a stand-in. In June 2022, Wayback static parses recover 92.2% of HTTP Archive's ad-tracker presence on the 410 jointly measured domains (Table C.10), which licenses the smaller Wayback fill in Section C.5.

Table C.9. HTTP Archive versus Blacklight, contemporaneous agreement (January 2025)

Measure pair	N	HA prev. (%)	BL prev. (%)	Agreement (%)	ρ
Ad trackers (same construct)	7,420	89.3	80.7	85.6	0.61
Any known tracker (HA) vs ad trackers (BL)	7,420	95.4	80.7	82.7	0.56
3p cookies: header-set (HA) vs all (BL)	2,602	84.0	67.3	74.2	0.52
FB requests (HA) vs FB pixel (BL)	7,420	43.3	26.9	78.3	0.57
GA/GTM presence (HA) vs GA remarketing (BL)	7,420	82.9	4.2	21.0	0.07

Note: Jointly measured domains, mobile emulation on both sides. Row labels state construct differences explicitly.

Table C.10. Wayback versus HTTP Archive, contemporaneous agreement (June 2022)

Measure	N	WB prev. (%)	HA prev. (%)	Agreement (%)	Recall of HA (%)	ρ
Ad trackers (static subset vs full request map)	410	89.3	96.6	92.2	92.2	0.37
Any known tracker	410	94.9	100.0	94.9	94.9	0.45
Facebook requests	410	36.6	56.3	59.8	46.8	0.24
Google Analytics/GTM	410	70.0	91.2	76.8	75.7	0.40
Any third party	410	96.6	100.0	96.6	96.6	0.44

Note: Wayback parses are static request maps of archived snapshots; dynamic behaviors do not execute in replay, so recall of HTTP Archive positives is the operative statistic.

C.5 Bounding the Coverage Gap

The main estimates set unscanned visits to zero. Here we replace the zero with fills and report the resulting interval for each mean exposure rate (Table C.11, Figure C.3). The first three columns are assumption-only fills: zero (the published construction), the scanned visit-weighted mean, and the scanned 90th percentile. The next columns use measurement:

HTTP Archive's June 2022 request maps cover 63.3% of the missing visit mass and Wayback snapshots another 2.4%; for covered visits the fill is E[Blacklight count | auxiliary presence] estimated on jointly measured domains, so magnitudes stay on the Blacklight scale, and only the residual missing mass takes the assumption-based fill. The calibrated interval puts the ad-tracker rate in $[6.19, 8.04]$ per visit against the published 5.19, and third-party cookies in $[7.09, 10.51]$ against 6.32. The direct audit of Section C.1 agrees from the other side. We keep the published tables as the defensible floor and report the bounds here.

Table C.11. Mean exposure rates under alternative fills for unscanned visits

	Assumption-only fill			HA-calibrated		HA + Wayback calibrated		
Measure	Zero	Mean	p90	Zero	p90	Zero	Mean	p90
Ad trackers	5.186	6.638	10.833	6.172	8.130	6.193	6.668	8.042
Third-party cookies	6.317	8.018	12.642	7.086	10.513	7.086	8.008	10.513
Facebook Pixel	0.086	0.107	0.086	0.098	0.098	0.099	0.105	0.099
Google Analytics (Remarketing)	0.010	0.012	0.010	0.011	0.011	0.011	0.012	0.011

Note: "Zero" reproduces the published construction. Calibrated columns fill auxiliary-covered visits with E[Blacklight count | auxiliary presence] from jointly measured domains; the residual missing mass takes the column's assumption fill. Facebook Pixel and Google Analytics rely on presence constructs with weak cross-instrument agreement (Section C.4) and are shown for completeness.

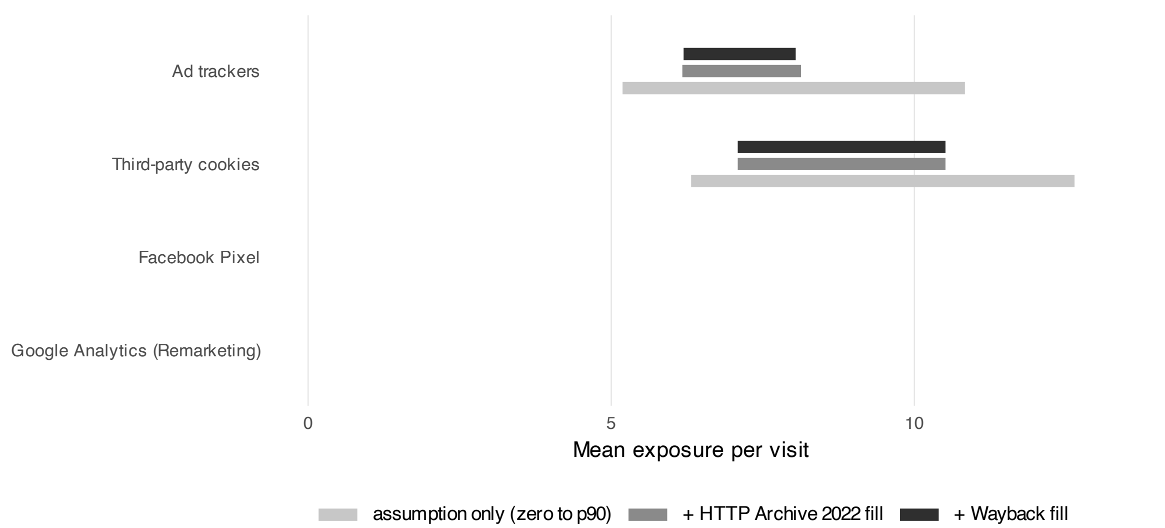


Figure C.3. Bounds on mean exposure rates under assumption-only and measurement-calibrated fills for unscanned visits.

C.6 Do the Threats Move the Comparisons?

The strands above bound levels. This section tests whether either threat moves the paper's comparisons.

Coverage is demographically flat. Regressing per-user scan coverage (Section C.2) on the demographic specification of Equation (6) yields $R^2 = 0.01$ with no coefficient significant at the five percent level; the largest is Race: Asian at 3.12 percentage points ($p = .28$) (Table C.12). Differential missingness cannot manufacture or mask group gaps of the size we study.

Table C.12. Per-user scan coverage (percentage points) by demographics

	Coverage (pp)
Woman	-0.90 (0.88)
Race: African American	-0.60 (1.39)
Race: Asian	3.12 (2.91)
Race: Hispanic	-0.70 (1.33)
Race: Other	-0.99 (2.04)
Educ: Some college	-0.05 (1.12)
Educ: College	0.78 (1.22)
Educ: Postgraduate	1.12 (1.49)
Age: 25–34	-0.36 (1.98)
Age: 35–49	-0.90 (1.77)
Age: 50–64	-0.96 (1.69)
Age: 65+	1.17 (1.71)

Note: OLS with Huber-White standard errors in parentheses. $R^2 = 0.01$, $n = 1{,}134$. * $p < .1$, ** $p < .05$, *** $p < .01$.

The demographic estimates are invariant to imputation. Re-estimating the exposure-rate regressions with unscanned visits imputed under each calibrated scenario leaves every conclusion intact (Table C.13, Figure C.4). The college gap stays small and marginal (ads 0.48 under zero-fill against 0.43 to 0.44 under measured fills, never significant at five percent under the latter); the 65+ coefficient is essentially unchanged (ads 2.89 against 2.80 to 2.82; cookies 3.18 against 3.09 to 3.14, all $p < .01$); and the Asian coefficient strengthens slightly (ads -1.24 to -1.44; cookies -1.61 to -1.86, all $p < .01$).

Timing is not neutral for the age comparison. Estimating the same demographic specification on same-instrument exposure rates at each measurement date shows that tracking growth between June 2022 and June 2025 concentrated on the sites older users frequent: the 65+ drift coefficient is +3.34 ad trackers and +7.33 cookies per visit (both $p < .01$) (Table C.14). The age gradient is large and significant under contemporaneous June 2022 measurement as well (+8.26 ad trackers per visit), but June 2025 measurement overstates the June 2022 gap by about two-fifths for ad trackers and more than doubles it for cookies.

Table C.13. Exposure-rate regressions under alternative fills for unscanned visits

	Ad trackers / visit				Third-party cookies / visit			
	Zero	Mean	HA	HA+WB	Zero	Mean	HA	HA+WB
Woman	-0.11 (0.22)	-0.05 (0.22)	-0.05 (0.22)	-0.04 (0.22)	0.01 (0.30)	0.08 (0.30)	0.08 (0.30)	0.08 (0.30)
Race: African American	-0.04 (0.36)	-0.00 (0.35)	-0.03 (0.35)	-0.04 (0.35)	-0.47 (0.45)	-0.43 (0.45)	-0.47 (0.45)	-0.47 (0.45)
Race: Asian	-1.24*** (0.29)	-1.44*** (0.31)	-1.43*** (0.31)	-1.44*** (0.31)	-1.61*** (0.43)	-1.84*** (0.46)	-1.86*** (0.46)	-1.86*** (0.46)
Race: Hispanic	0.15 (0.34)	0.19 (0.33)	0.21 (0.33)	0.21 (0.33)	0.12 (0.47)	0.17 (0.47)	0.19 (0.47)	0.19 (0.47)
Race: Other	-0.23 (0.44)	-0.17 (0.44)	-0.16 (0.45)	-0.15 (0.45)	0.10 (0.67)	0.17 (0.68)	0.16 (0.68)	0.16 (0.68)
Educ: Some college	0.21 (0.27)	0.21 (0.27)	0.23 (0.27)	0.22 (0.27)	0.22 (0.38)	0.23 (0.38)	0.21 (0.38)	0.21 (0.38)
Educ: College	0.48 (0.30)	0.43 (0.30)	0.43 (0.30)	0.44 (0.30)	0.71* (0.43)	0.66 (0.42)	0.65 (0.42)	0.65 (0.42)
Educ: Postgraduate	0.30 (0.34)	0.23 (0.33)	0.23 (0.33)	0.23 (0.33)	0.38 (0.47)	0.30 (0.46)	0.29 (0.46)	0.29 (0.46)
Age: 25–34	0.51* (0.29)	0.53* (0.30)	0.54* (0.30)	0.52* (0.30)	0.64 (0.43)	0.67 (0.44)	0.71 (0.44)	0.71 (0.44)
Age: 35–49	1.91*** (0.32)	1.97*** (0.33)	1.96*** (0.33)	1.95*** (0.33)	1.98*** (0.48)	2.05*** (0.49)	2.09*** (0.49)	2.09*** (0.49)
Age: 50–64	2.16*** (0.33)	2.22*** (0.33)	2.22*** (0.33)	2.21*** (0.33)	2.48*** (0.44)	2.55*** (0.45)	2.59*** (0.45)	2.59*** (0.45)
Age: 65+	2.89*** (0.33)	2.82*** (0.34)	2.81*** (0.34)	2.80*** (0.34)	3.18*** (0.45)	3.09*** (0.47)	3.14*** (0.47)	3.14*** (0.47)

Note: Each column re-estimates Equation (6) on the per-user rate under the named fill for unscanned visits. "Zero" reproduces the published estimates. Huber-White standard errors in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

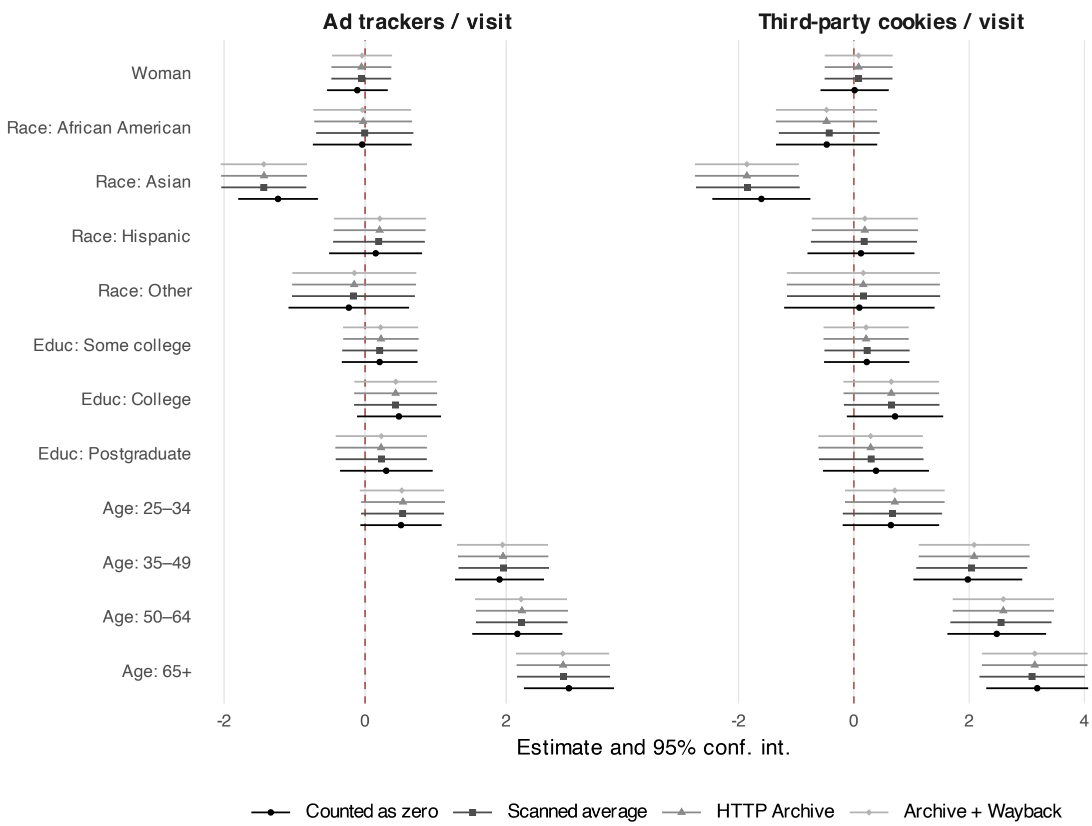


Figure C.4. Key demographic coefficients under each fill scenario for unscanned visits.

The qualitative conclusion holds at either date; magnitudes should be read as of the scan date. The widening itself is substantively interesting: the age skew in tracking exposure grew between 2022 and 2025. Education gaps show no differential drift (College: -0.12 ads per visit, $p = .86$), and neither does the Asian coefficient (-0.66, $p = .21$).

Table C.14. Same-instrument demographic gaps at both measurement dates (HTTP Archive)

	Ad trackers / visit			Cookies (header-set) / visit		
	June 2022	June 2025	Drift	June 2022	June 2025	Drift
Woman	-0.13 (0.76)	-0.96 (0.85)	-0.83* (0.50)	-0.47 (0.71)	-1.50 (1.07)	-1.04 (0.72)
Race: African American	-2.01** (0.98)	-2.43** (1.13)	-0.41 (0.76)	-1.49* (0.87)	-0.77 (1.57)	0.72 (1.14)
Race: Asian	-0.54 (2.94)	-1.20 (3.07)	-0.66 (0.52)	0.03 (3.05)	-1.43 (4.14)	-1.47 (1.30)
Race: Hispanic	-0.70 (1.11)	0.07 (1.34)	0.77 (0.83)	0.37 (1.07)	1.21 (1.72)	0.84 (1.21)
Race: Other	0.35 (1.80)	-1.86 (1.69)	-2.21 (1.58)	1.14 (1.61)	-1.29 (2.01)	-2.43 (1.56)
Educ: Some college	-0.28 (0.90)	-0.90 (1.00)	-0.62 (0.60)	0.33 (0.81)	-0.51 (1.21)	-0.85 (0.84)
Educ: College	1.42 (1.09)	1.30 (1.27)	-0.12 (0.69)	1.19 (0.97)	1.83 (1.59)	0.64 (1.05)
Educ: Postgraduate	1.22 (1.12)	-0.26 (1.32)	-1.48 (0.91)	0.31 (1.00)	-1.56 (1.62)	-1.87 (1.25)
Age: 25–34	2.46 (1.63)	3.13* (1.61)	0.67 (0.61)	1.96 (1.43)	3.22* (1.75)	1.26 (0.97)
Age: 35–49	3.37*** (1.10)	5.28*** (1.21)	1.91*** (0.74)	3.21*** (1.04)	7.55*** (1.59)	4.34*** (1.09)
Age: 50–64	5.31*** (1.12)	8.03*** (1.21)	2.72*** (0.68)	4.16*** (1.04)	10.08*** (1.55)	5.92*** (1.03)
Age: 65+	8.26*** (1.19)	11.60*** (1.28)	3.34*** (0.77)	6.35*** (1.20)	13.68*** (1.67)	7.33*** (1.10)

Note: Each pair of date columns estimates Equation (6) on per-user rates built from the same browsing data and the same instrument at that date; the drift column regresses the within-user difference. Huber-White standard errors in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

Google's reach extends to the unscanned web. On the fresh July 2026 scans of the audit sample, 68.3% of visits-stratum domains (CI $[56, 78]$) and 72.1% of uniform-stratum domains (CI $[60, 81]$) load at least one Google-owned third party, statistically indistinguishable from the 69.9% visit-weighted share in the scanned population (Table C.15). The concentration finding of Section 4.3 extends to the domains we could not scan.

Table C.15. Google third-party presence on scanned versus unscanned domains

Set	Share (%)	N	95% CI
Scanned population, Jan 2025 (visit-weighted)	69.9	34,512	–
Unscanned sample, visits stratum (scanned Jul 2026)	68.3	63	[56, 78]
Unscanned sample, uniform stratum (scanned Jul 2026)	72.1	68	[60, 81]

Benchmarks for the gaps. Table C.16 re-expresses the key exposure-rate gaps relative to the reference group's mean rate and as Cohen's d, the figures used in Section 5.

C.7 Desktop-Only Sensitivity

Metering is more complete on desktop than on mobile, where a share of activity moves through applications the extension never sees. Restricting the sample to the 673 panelists whose logged activity is desktop only leaves the pattern intact (Table C.17). Asian panelists remain less exposed per visit on five of the seven measures, the age gradient survives at close to full size for ad trackers (3.472 against 3.555) and third-party cookies (3.524 against 3.926), and the gender gap in canvas fingerprinting is if anything larger (0.021 against 0.017).

Table C.16. Demographic gaps in exposure rates as shares of the reference mean and effect sizes

Measure	Group	Coef.	Ref. mean	% of ref.	Cohen's d
Ad trackers / visit	Woman	-0.11	5.20	-2	-0.03
Ad trackers / visit	Educ: College	0.48	5.03	+10	0.13
Ad trackers / visit	Educ: Postgraduate	0.30	5.03	+6	0.08
Ad trackers / visit	Age: 65+	2.89***	3.32	+87	0.77
Ad trackers / visit	Race: Asian	-1.24***	5.37	-23	-0.33
Third-party cookies / visit	Woman	0.01	6.27	+0	0.00
Third-party cookies / visit	Educ: College	0.71*	6.08	+12	0.14
Third-party cookies / visit	Educ: Postgraduate	0.38	6.08	+6	0.07
Third-party cookies / visit	Age: 65+	3.18***	4.19	+76	0.61
Third-party cookies / visit	Race: Asian	-1.61***	6.58	-24	-0.31

Note: Coefficients from the published exposure-rate regressions (zero-fill). * $p < .1$, *** $p < .01$.

Two results weaken: the 65+ coefficient on canvas fingerprinting loses significance, and the education coefficients, already small once normalized, move without a consistent sign. Halving the sample costs precision, and the desktop-only estimates should be read as a check on sign and rough magnitude rather than as a second set of point estimates. Rates in this check are computed per scanned visit for both samples, so magnitudes are not directly comparable to Table D.2, which normalizes by all visits.

Table C.17. Demographic gaps among desktop-only panelists

	Woman		Race: Asian		Educ: College		Age: 65+	
Measure	All	Desktop	All	Desktop	All	Desktop	All	Desktop
Ad Trackers	-0.052	-0.133	-1.867***	-1.590***	0.595	0.700	3.555***	3.472***
Third-Party Cookies	0.088	0.012	-2.381***	-1.688***	0.959*	0.684	3.926***	3.524***
Facebook Pixel	0.010	0.001	-0.031**	-0.016	-0.013	0.008	0.014	0.004
Google Analytics (Remarketing)	0.000	-0.004	-0.004	-0.003	-0.002	-0.000	-0.010	-0.010
Keylogging	0.002	0.006	-0.023***	-0.029***	0.007	0.002	0.045***	0.046***
Session Recording	0.012**	0.000	-0.004	-0.026***	-0.004	0.008	0.035***	0.022**
Canvas Fingerprinting	0.017**	0.021**	-0.026***	-0.032***	0.005	-0.002	0.030**	0.018

Note: OLS with Huber-White (HC1) standard errors, the specification of Equation (6). "All" is every panelist with a metered device, "Desktop" the 673 whose logged activity is desktop only. Rates are per *scanned* visit in both columns, so they are not comparable to Table D.2, which divides by all visits. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

D Regression Tables

Estimates displayed in Figures 2 and 3.

Table D.1. Demographic differences in cumulative tracking exposure. OLS with Huber-White (HC1) standard errors in parentheses. Ad trackers, third-party cookies and top-organisation visits are expressed in hundreds. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Measure	Ad (00s)	Cookies (00s)	FB Pixel	GA Remkt.	Keylogger	Session rec	Canvas FP	Top org visits (00s)
Woman	-25	-18	-23	1.6	-6.9	-6.5	74***	-3.8**
	(19)	(21)	(28)	(5.0)	(41)	(15)	(29)	(1.9)
Race: African American	-20	-25	-29	-5.4	-27	-6.4	-44	-3.4
	(27)	(29)	(42)	(6.9)	(58)	(21)	(37)	(2.6)
Race: Asian	17	31	22	28	-84	-49***	17	11
	(51)	(59)	(62)	(27)	(66)	(19)	(63)	(7.4)
Race: Hispanic	-21	-21	-56**	-15***	-27	-4.0	-11	-1.2
	(24)	(26)	(28)	(4.5)	(61)	(18)	(43)	(2.8)
Race: Other	6.7	22	10.0	-11*	-100**	-19	178	-2.5
	(43)	(56)	(74)	(6.6)	(50)	(23)	(122)	(3.6)
Educ: Some college	13	25	46	9.1	2.8	-8.5	28	3.7*
	(20)	(24)	(35)	(5.7)	(48)	(16)	(35)	(2.2)
Educ: College	81***	104***	46	11**	112*	48*	46	8.8***
	(29)	(33)	(34)	(5.3)	(61)	(25)	(38)	(2.6)
Educ: Postgraduate	65*	68*	115*	12	7.4	42*	118*	8.1**
	(35)	(36)	(62)	(12)	(55)	(25)	(67)	(3.3)
Age: 25–34	26	29	57	-10.0	-75	36*	44	1.9
	(17)	(22)	(38)	(11)	(100)	(21)	(43)	(3.7)
Age: 35–49	98***	96***	95***	6.5	96	61***	75**	3.5
	(24)	(27)	(35)	(11)	(109)	(21)	(31)	(3.5)
Age: 50–64	136***	156***	144***	-8.9	82	86***	124***	4.6
	(25)	(29)	(38)	(9.5)	(108)	(25)	(33)	(3.4)
Age: 65+	208***	238***	243***	-3.3	163	126***	247***	7.9**
	(24)	(29)	(43)	(9.7)	(107)	(23)	(42)	(3.3)
Dependent variable mean	197	231	285	27	212	140	236	22
R^2	0.061	0.063	0.037	0.019	0.021	0.036	0.045	0.032
Observations	1,134	1,134	1,134	1,134	1,134	1,134	1,134	1,134

Table D.2. Demographic differences in tracking exposure per visit. OLS with Huber-White (HC1) standard errors in parentheses. Reference categories are male, white, high school or below, and under 25. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Measure	Ad	Cookies	FB Pixel	GA Remkt.	Keylogger	Session rec	Canvas FP	Top org share
Woman	-0.110	0.012	0.004	0.000	0.000	0.008^{**}	0.013^{**}	-0.010
	(0.220)	(0.302)	(0.005)	(0.002)	(0.004)	(0.004)	(0.006)	(0.010)
Race: African American	-0.041	-0.472	0.003	0.000	0.004	0.011	0.000	-0.013
	(0.358)	(0.448)	(0.010)	(0.003)	(0.007)	(0.008)	(0.009)	(0.016)
Race: Asian	-1.24^{***}	-1.61^{***}	-0.020^{*}	-0.003	-0.017^{***}	0.000	-0.020^{**}	0.009
	(0.288)	(0.434)	(0.010)	(0.002)	(0.005)	(0.015)	(0.008)	(0.030)
Race: Hispanic	0.150	0.122	0.002	-0.001	0.004	0.007	0.003	0.008
	(0.337)	(0.473)	(0.008)	(0.004)	(0.008)	(0.005)	(0.007)	(0.015)
Race: Other	-0.231	0.095	-0.007	-0.004^{**}	-0.010	0.000	0.011	-0.004
	(0.436)	(0.665)	(0.008)	(0.002)	(0.008)	(0.007)	(0.012)	(0.023)
Educ: Some college	0.206	0.223	0.000	0.000	0.000	-0.005	0.008	0.018
	(0.275)	(0.377)	(0.007)	(0.003)	(0.005)	(0.005)	(0.007)	(0.013)
Educ: College	0.480	0.714^{*}	-0.010	-0.003	0.006	-0.003	0.004	0.034^{**}
	(0.304)	(0.427)	(0.007)	(0.003)	(0.007)	(0.006)	(0.007)	(0.014)
Educ: Postgraduate	0.300	0.384	-0.007	-0.005	0.000	-0.002	0.020^{*}	0.052^{***}
	(0.336)	(0.469)	(0.008)	(0.003)	(0.008)	(0.007)	(0.011)	(0.017)
Age: 25–34	0.509^{*}	0.642	-0.007	-0.007	0.002	0.013^{*}	0.001	-0.026
	(0.294)	(0.428)	(0.013)	(0.005)	(0.007)	(0.007)	(0.010)	(0.022)
Age: 35–49	1.91^{***}	1.98^{***}	0.012	0.001	0.022^{***}	0.022^{***}	0.010	-0.029
	(0.322)	(0.482)	(0.014)	(0.006)	(0.008)	(0.007)	(0.010)	(0.020)
Age: 50–64	2.16^{***}	2.48^{***}	0.007	-0.005	0.027^{***}	0.026^{***}	0.009	-0.059^{***}
	(0.326)	(0.437)	(0.013)	(0.005)	(0.008)	(0.007)	(0.009)	(0.019)
Age: 65+	2.89^{***}	3.18^{***}	0.015	-0.007	0.036^{***}	0.028^{***}	0.027^{***}	-0.052^{***}
	(0.327)	(0.451)	(0.013)	(0.005)	(0.008)	(0.007)	(0.010)	(0.020)
Dependent variable mean	5.19	6.32	0.086	0.010	0.041	0.049	0.066	0.585
R^2	0.072	0.049	0.013	0.010	0.033	0.023	0.023	0.024
Observations	1,134	1,134	1,134	1,134	1,134	1,134	1,134	1,134

E Holding browsing volume constant without assuming proportionality

Equation (6) holds browsing volume constant by dividing cumulative exposure by visits, which imposes proportionality between the two. Equation (7) does not impose it: browsing volume and its square enter the right-hand side of a median regression on cumulative exposure, and the data decide the shape. Studies of other online risks adjust this second way, so reporting both is what allows a difference between our demographic coefficients and theirs to be read as a difference in the phenomenon rather than in the arithmetic.

Table E.1 gives both specifications for all seven tracking measures and for the visits observed by the top organization. Panel A is demographics alone; Panel B adds the volume terms. The comparison the text draws is vertical: what happens to a coefficient when volume is allowed in.

Two features of the estimation are worth stating. Standard errors are bootstrapped over panelists, 1,000 draws, and every draw converged. And because `statsmodels` solves the median regression by iteratively reweighted least squares — an approximation to what is exactly a linear program — each reported fit is re-solved through an exact linear-programming solver and the two objectives are required to agree; the largest relative gap across the sixteen fits is 5×10^{-9}.

The education result reproduces what other-risk studies report: the college coefficient falls from 5,559 to 68 and becomes indistinguishable from zero. The age result does not: the 65+ coefficient falls but remains large and significant. Browsing volume enters with a large positive linear term and a negative quadratic one, the concave shape reported elsewhere, though here the quadratic is not precisely estimated for any measure.

Table E.1. Demographic differences in cumulative exposure, before and after adjusting for browsing volume

	Ad	Cookies	FB Pixel	GA	Keylogger	Session rec	Canvas FP	Top org visits
Panel A. Demographics only								
Woman	-788 (798)	-374 (989)	9 (14)	2** (1)	1 (2)	5 (7)	9 (8)	-113 (113)
Race: African American	1,245 (1,141)	690 (1,456)	-1 (18)	-2** (1)	1 (4)	5 (9)	-4 (11)	58 (145)
Race: Asian	3,167 (1,996)	3,013 (2,733)	81 (51)	2 (5)	1 (6)	2 (13)	15 (16)	805** (351)
Race: Hispanic	405 (963)	-175 (1,135)	15 (18)	-1 (1)	-2 (2)	6 (10)	-4 (10)	-4 (168)
Race: Other	-223 (2,180)	-2,434 (2,952)	-16 (29)	-2 (1)	-3 (4)	2 (14)	-5 (21)	-45 (304)
Educ: Some college	2,519*** (921)	3,017** (1,213)	36** (16)	2* (1)	4 (3)	11 (7)	26** (10)	362*** (115)
Educ: College	5,559*** (1,254)	6,193*** (1,668)	60*** (20)	3** (1)	10** (4)	23** (10)	56*** (14)	816*** (199)
Educ: Postgraduate	7,826*** (1,973)	8,543*** (2,758)	60* (36)	2 (2)	13* (8)	32* (18)	48*** (17)	692*** (260)
Age: 25–34	927 (899)	-117 (1,132)	5 (15)	-1 (1)	2 (2)	12 (8)	-4 (11)	49 (159)
Age: 35–49	1,809* (942)	1,281 (1,105)	39** (17)	1 (2)	6** (2)	29*** (8)	8 (10)	143 (159)
Age: 50–64	4,325*** (1,422)	5,162*** (1,737)	62** (27)	2 (2)	25*** (4)	56*** (12)	36*** (13)	264 (192)
Age: 65+	13,565*** (2,210)	14,331*** (2,620)	164*** (32)	6*** (2)	42*** (12)	113*** (13)	92*** (22)	956*** (200)
Panel B. Demographics and browsing volume								
Woman	254 (195)	194 (226)	4 (4)	1** (0)	0 (2)	5 (3)	5* (3)	-13 (11)
Race: African American	234 (259)	-20 (289)	-3 (6)	-1* (0)	-0 (2)	-1 (4)	-4 (4)	-7 (13)
Race: Asian	-1,024 (657)	-397 (737)	-3 (13)	0 (2)	-4 (7)	-9 (10)	-1 (11)	22 (65)
Race: Hispanic	21 (254)	-182 (276)	-1 (4)	-1 (0)	-2 (2)	6 (4)	2 (3)	-2 (13)
Race: Other	-124 (333)	-165 (438)	-1 (8)	-1 (1)	-2 (3)	-6 (8)	2 (9)	-19 (28)
Educ: Some college	-86 (210)	-125 (235)	2 (5)	0 (1)	1 (2)	-3 (3)	6* (3)	13 (13)
Educ: College	68 (292)	331 (293)	0 (5)	0 (1)	-1 (2)	-3 (5)	1 (4)	30 (20)
Educ: Postgraduate	440 (391)	341 (419)	1 (7)	-0 (1)	-0 (3)	1 (7)	3 (5)	38 (25)
Age: 25–34	325 (405)	72 (459)	7 (7)	-0 (1)	1 (2)	8* (4)	-0 (4)	-10 (20)
Age: 35–49	1,108*** (373)	828** (421)	8 (7)	1 (1)	6** (2)	15*** (5)	4 (3)	-10 (17)
Age: 50–64	1,203*** (377)	1,184** (466)	8 (7)	-0 (1)	8*** (3)	19*** (5)	6 (4)	-29 (20)
Age: 65+	2,427*** (672)	1,957*** (699)	16 (11)	1 (1)	10* (5)	36*** (9)	17** (7)	-27 (24)
Total visits (scaled)	220,518*** (14,364)	265,339*** (15,332)	3,426*** (213)	110*** (23)	759*** (160)	1,421*** (113)	2,152*** (190)	26,881*** (950)
Total visits^2 (scaled)	-135,413** (60,127)	-136,784*** (51,690)	-3,218*** (615)	86 (84)	-449 (795)	-1,377*** (346)	-2,126*** (415)	-4,207 (4,347)

Note: Median regressions ($\tau = 0.5$) of cumulative exposure, with standard errors from 1,000 bootstrap draws over panelists in parentheses. Panel A estimates Equation (6) at the median; Panel B estimates Equation (7), adding total visits and their square, each rescaled to the unit interval. Reference categories are male, white, high school or below, and under 25. * $p < .1$, ** $p < .05$, *** $p < .01$.

F Robustness of the age gap to the rate denominator

The exposure rate divides a panelist's tracker encounters by that panelist's own visits, and the demographic models then average those ratios with equal weight across people. Two features of the panel put pressure on that construction. The distribution of visits is long-tailed, from a minimum of 1 to a maximum of 46,563, so the precision of the ratio varies enormously across panelists; and 19 panelists browsed fewer than ten times, 90 fewer than a hundred. A rate computed from two visits is close to noise, and equal weighting gives it the same standing as a rate computed from tens of thousands.

Table F.1 re-estimates Equation (6) under progressively stricter minimum-visit restrictions and, in the last row, under weights proportional to each panelist's visits. The weighted row is not a robustness check in the usual sense but a different estimand: it describes the average *visit* rather than the average *person*. We report the unweighted person-level estimate in the main text because the paper's claims are about people.

Every alternative gives a larger 65+ coefficient than the published one, for both headline measures. The direction is what makes the check informative. The panelists with the least reliable denominators are disproportionately younger and are pulling the estimate down, so the published specification is the conservative member of the family rather than a fortunate choice within it.

Table F.1. Age gap (65+ vs. under 25) under alternative treatments of the rate denominator

		Ad trackers		Third-party cookies	
Specification	n	Coef.	SE	Coef.	SE
Published (all panelists)	1,134	2.891***	(0.327)	3.181***	(0.451)
Drop < 10 visits	1,115	2.949***	(0.327)	3.257***	(0.452)
Drop < 50 visits	1,075	2.997***	(0.323)	3.360***	(0.430)
Drop < 100 visits	1,044	3.024***	(0.324)	3.370***	(0.428)
Drop < 250 visits	967	3.082***	(0.319)	3.308***	(0.399)
Visit-weighted	1,134	3.349***	(0.338)	3.634***	(0.404)

Note: Each cell is the 65+ coefficient from Equation (6) estimated on the stated subsample, with Huber-White (HC1) standard errors. The final row weights panelists by their total visits and therefore estimates the gap for the average visit rather than the average person. * $p < .1$, ** $p < .05$, *** $p < .01$.

Two further checks bear on the age result specifically. The paper bins age, so the gradient could in principle be an artifact of where the bins were cut. Table F.2 fits birth year flexibly instead, with a natural cubic spline and the same covariates. The distinction that matters is between *any* birth-year effect and a *nonlinear* one: a spline basis already contains a straight line, so a joint test of its columns cannot separate them. We therefore residualise the basis on the linear term and test what remains. Birth year matters for every measure the paper headlines, and nonlinearity is significant for none of them — the gradient runs essentially straight across cohorts, which is what makes the binning innocuous rather than load-bearing. The penalised-smooth columns, fit with `mgcv`, agree.

Older and younger panelists also do not use the same devices, and mobile and desktop browsing are tracked differently, so the age gap could be a device effect in demographic clothing. The clean test would be within-person, and this panel cannot support it: 1,108 of the 1,134 panelists browse on exactly one device type. Table F.3 gives what the data does support — the gradient within each device group, and device added to Equation (6). The coefficient barely moves and holds within both groups. None of these identifies a device effect free of selection; together they establish that device composition is not manufacturing the gap.

Table F.2. Flexible tests of the birth-year gradient. The first two columns are OLS with HC1 errors, testing any birth-year effect and the nonlinear basis alone; the last two are the equivalent penalised smooth.

Measure	Any age effect	Nonlinearity	Smooth EDF	Smooth p
Ad Trackers	¡ .001	.094	1.89	¡ .001
Third-Party Cookies	¡ .001	.443	1.45	¡ .001
Facebook Pixel	.045	.185	1.00	.081
Google Analytics (Remarketing)	.172	.364	1.00	.094
Keylogging	¡ .001	.327	1.68	¡ .001
Session Recording	.013	.469	1.54	.005
Canvas Fingerprinting	.054	.528	1.26	.005
Top organization share	.002	.110	1.66	.002

Table F.3. The 65+ coefficient from Equation (6) on exposure per visit, pooled, with device added, and within each metered device group. *** $p < .01$, ** $p < .05$, * $p < .1$.

Measure	Pooled	+ Device	Desktop only	Mobile only
Ad Trackers	2.892***	2.897***	2.864***	2.845***
Third-Party Cookies	3.182***	3.223***	2.994***	3.757***
Canvas Fingerprinting	0.027***	0.023**	0.018	0.030***
Session Recording	0.028***	0.030***	0.017**	0.056***
Keylogging	0.036***	0.033***	0.040***	0.024**

G Where the age gap comes from: category mix or site choice

That older panelists meet more trackers per visit is a fact without a mechanism, and two readings fit it. Older panelists might spend their time in categories that are more heavily tracked, in which case the gap is about *what* they browse. Or, within the same categories, they might land on more heavily tracked sites, in which case it is about *which* sites they choose.

Writing a group's exposure as a sum over categories of (share of visits) × (tracking per visit in that category), the gap between two groups splits into a composition term (different category mix, holding tracking intensity fixed), a site-choice term (different tracking within the same categories, holding mix fixed), and the residual cross term. We prefer this to a regression-based decomposition because it reads directly off observed quantities and sidesteps the choice of reference coefficients. Both base groups are computed, since the split is not symmetric in which group supplies the weights; Table G.1 reports the under-25 base.

Site choice dominates. Of the 3.66 ad-tracker gap, 3.00 is site choice against 0.26 composition, and the pattern repeats for third-party cookies (3.20 against 0.48) and canvas fingerprinting. Older panelists are not mostly visiting different *kinds* of site; within the same kinds, they are visiting more heavily tracked ones.

Three limits. Categories are the meter's own per-visit labels, not a considered taxonomy, and 10.3% of visits carry none; those form their own bucket rather than being dropped. Visits carrying several labels are split fractionally across them so shares sum to one. And a shift-share describes an association, not a choice process: it says where in the browsing the gap sits, not why anyone browses as they do.

Table G.1. Decomposition of the 65+ versus under-25 gap in exposure per visit, under-25 base. Brackets are 95% percentile intervals from 500 resamples of panelists.

Measure	Total gap	Composition	Site choice	Interaction
Ad Trackers	3.657	0.256 [0.068, 0.576]	2.995 [2.251, 3.734]	0.407
Third-Party Cookies	4.006	0.484 [0.280, 0.891]	3.195 [2.372, 4.100]	0.327
Canvas Fingerprinting	0.043	0.007 [0.001, 0.014]	0.029 [0.012, 0.045]	0.008
Session Recording	0.022	0.007 [0.002, 0.014]	0.014 [0.001, 0.026]	0.001

H What today's blocklists would leave behind

Each Blacklight scan names the third-party domains responsible for each behavior it detects. That makes it possible to apply a blocklist to those domains and recompute every measure as arithmetic rather than simulation: we match the 444,485 domain–script pairs against EasyList, EasyPrivacy and Disconnect, pinned by commit and checksum, and recompute exposure with the blocked third parties removed.

Two caveats govern everything in this section. Blacklight stores hostnames, so a rule keyed on a URL path cannot fire; each residual is therefore reported as an interval, from blocking only what a domain rule certainly stops to crediting every host a list names. And the Facebook Pixel and Google Analytics cards name no responsible domain at all, so their attribution is assumed. Facebook Pixel's 100% survival in Table H.1 is an artifact of that, not a finding about the pixel.

Table H.1 gives the levels. Blocking removes most ad-tracker and third-party-cookie exposure but leaves 93.8% of canvas fingerprinting standing, which is what fingerprinting scripts are built to do: they are served from the first-party origin or from hosts the lists do not carry.

Table H.2 asks whether blocking narrows the age gap. Unadjusted, panelists 65+ meet 1.90× the ad-tracker exposure of under-25s; behind all three lists, 1.26×. But a blocklist that removes 86% of exposure will move the gap for arithmetic reasons alone, so Table H.3 compares the real list against 10,000 random sets of third parties calibrated to remove the same amount. The narrowing survives for third-party cookies ($p = .001$) and is marginal for ad trackers ($p = .05$). The apparent *widening* for session recording does not survive its placebo ($p = .18$) and we do not report it as evidence of regressive protection; only thirteen third parties carry that behavior, which makes both the estimate and the placebo distribution lumpy, and the placebo interval is correspondingly uninformative at its upper end.

Table H.4 recomputes the residual age effect under four age codings. The conclusion does not turn on the coding: ad trackers and third-party cookies retain about a tenth of their unblocked age effect under every continuous or binned specification. The final column reads differently by construction — where a measure reaches nearly everyone before and after blocking, a whether-at-all outcome has almost no variation left to explain.

Table H.5 addresses a different worry. Blacklight loads one logged-out page per domain, and 23.7% of successful scans return nothing on all seven measures while carrying 20.2% of panel visits. Treating those as genuine zeros is a choice. Excluding them, or imputing them at the mean of domains that did show tracking, raises every level and leaves every age coefficient significant in the same direction. No conclusion depends on the zero; the levels do, and they are understated because of it.

Finally, Table H.6 rakes the panel onto CPS 2022 margins. This is not a probability-sample estimate. It answers only: if the country browsed the way people with these demographics browsed, how many would have met each technique in a month? It corrects demographic composition and nothing else — not selection into a metered panel, and not the zero-fill above.

Table H.1. Exposure per visit surviving each blocklist tier. Brackets give the optimistic bound, crediting every host a list names. The final column is the share of unblocked exposure surviving all three lists.

Measure	Unblocked	EasyList	EasyPrivacy	All three	Survives (%)
Ad Trackers	5.186	0.915 [0.698]	3.474 [2.440]	1.222 [0.149]	23.6
Third-Party Cookie Domains	3.325	0.713 [0.633]	2.393 [1.602]	0.997 [0.270]	30.0
Canvas Fingerprinting	0.066	0.059 [0.057]	0.064 [0.045]	0.062 [0.041]	93.8
Session Recording	0.049	0.006 [0.005]	0.004 [0.003]	0.004 [0.003]	8.6
Keylogging	0.041	0.012 [0.012]	0.012 [0.007]	0.012 [0.007]	30.0
Facebook Pixel	0.086	0.000 [0.000]	0.086 [0.000]	0.086 [0.000]	100.0
Google Analytics (Remarketing)	0.010	0.010 [0.010]	0.010 [0.010]	0.000 [0.000]	0.0

Table H.2. The 65+ versus under-25 exposure ratio before and after blocking, with the change in log-ratio. Brackets are 95% percentile intervals from 1,000 resamples of panelists.

Measure	Ratio before	Ratio after	Δ log-ratio
Ad Trackers	1.90	1.26	-0.410 [-0.486, -0.341]
Third-Party Cookie Domains	1.96	1.18	-0.511 [-0.611, -0.411]
Canvas Fingerprinting	1.60	1.68	+0.050 [-0.016, +0.145]
Session Recording	1.99	5.78	+1.066 [+0.266, +2.062]
Keylogging	2.97	0.85	-1.254 [-1.493, -0.931]

Table H.3. Placebo test. Random sets of third parties are drawn to remove the same exposure as the real blocklist; the placebo column gives the median and 95% range of their gap shifts over 10,000 draws.

Measure	Exposure removed	Observed shift	Placebo	p
Ad Trackers	85.6%	-0.467	+0.059 [-0.523, +0.542]	0.054
Third-Party Cookies	75.3%	-0.463	+0.053 [-0.368, +0.283]	0.001
Canvas Fingerprinting	6.1%	+0.050	+0.010 [-0.055, +0.116]	0.370
Session Recording	91.4%	+1.066	-0.035 [-1.545, +24.625]	0.180
Keylogging	70.3%	-1.263	-0.965 [-2.173, +1.481]	0.511

Table H.4. Residual age effect as a percentage of the unblocked one, under four codings of age. 100 means blocking leaves the age gap where it found it.

Measure	65+ vs. < 25	65+ vs. rest	Per decade	Any exposure
Ad Trackers	9	10	10	100
Canvas Fingerprinting	103	106	105	95
Keylogging	-8	11	-4	121
Session Recording	19	22	16	194
Third-Party Cookie Domains	8	12	11	100

Table H.5. Exposure per visit under three treatments of scans returning nothing on all seven measures, with the 65+ coefficient from Equation (6) in parentheses. *** $p < .01$, ** $p < .05$, * $p < .1$.

Measure	Published (zero)	Excluded	Imputed at mean
Ad Trackers	5.186 (+2.891***)	6.431 (+3.816***)	6.932 (+3.260***)
Third-Party Cookie Domains	6.317 (+3.181***)	7.840 (+4.221***)	8.750 (+3.695***)
Canvas Fingerprinting	0.066 (+0.027***)	0.081 (+0.039***)	0.081 (+0.030***)
Session Recording	0.049 (+0.028***)	0.061 (+0.038***)	0.071 (+0.033***)
Keylogging	0.041 (+0.036***)	0.051 (+0.047***)	0.050 (+0.038***)

Table H.6. Share of adults encountering each technique at least once in the month, raked onto CPS 2022 margins, and the implied count of US adults. Brackets are 95% percentile intervals over 500 resamples, re-raking within each.

Measure	Sample (%)	Weighted (%)	US adults (m)	Behind a blocker (m)
Ad Trackers	99.6	99.5	254.1 [253.0, 255.0]	253.6 [252.3, 254.6]
Third-Party Cookies	99.4	99.3	253.5 [252.1, 254.6]	253.5 [252.1, 254.6]
Canvas Fingerprinting	92.2	91.9	234.6 [230.5, 238.5]	233.4 [229.2, 237.3]
Session Recording	91.4	91.2	232.7 [228.4, 236.8]	111.1 [103.9, 118.5]
Keylogging	85.4	85.0	217.1 [212.3, 222.9]	191.9 [185.3, 198.4]

I Organization tracking weighted by time

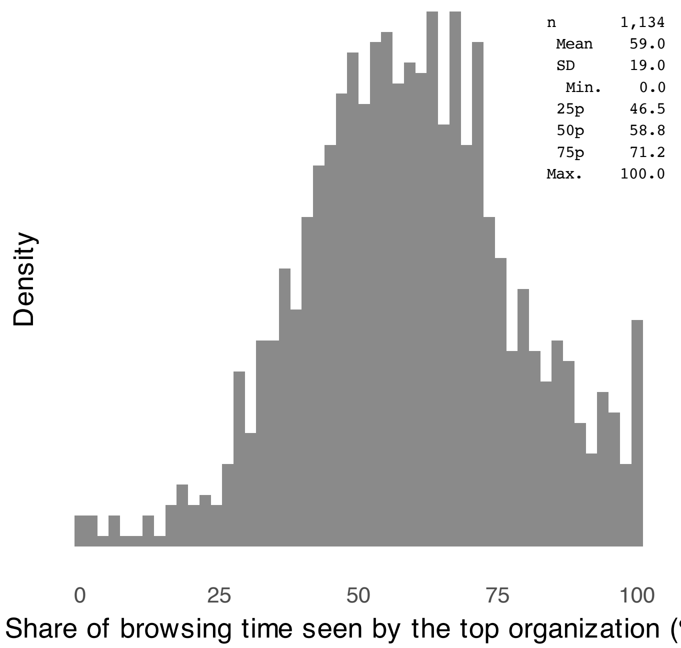


Figure I.1. The largest share of each user's browsing time online tracked by a single organization (Equation (5)):

$$\text{Tracking share}_{ij}^{(\text{dur})} = \frac{\sum_{v \in \mathcal{V}_i} \mathbf{1}(j \in O_{iv}) \cdot t_{iv}}{\sum_{v \in \mathcal{V}_i} t_{iv}}.$$

See Figure 5d for the corresponding figure for tracking shares by site visits.

J Top Tracking Domains

Table J.1. Top domains contributing to exposure

	Ads	Cookies	FB Pixel	GA	Session rec	Keyloggers	Canvas FP
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1	yahoo.com (169k)	yahoo.com (169k)	ebay.com (22k)	kohls.com (1.9k)	yougov.com (33k)	yahoo.com (169k)	live.com (59k)
2	google.com (689k)	google.com (689k)	capitaloneshopping.com (12k)	pixiv.net (1.5k)	capitalone.com (7.9k)	capitaloneshopping.com (12k)	microsoft.com (15k)
3	live.com (59k)	live.com (59k)	chase.com (11k)	force.com (1.3k)	xfinity.com (6.9k)	smugmug.com (10k)	linkedin.com (13k)
4	aol.com (27k)	bing.com (134k)	hulu.com (8.9k)	mheducation.com (1.3k)	att.com (3.9k)	weather.com (2.5k)	capitaloneshopping.com (12k)
5	microsoft.com (15k)	microsoft.com (15k)	nielseniq.com (8.5k)	tupperware.com (1.3k)	cbssports.com (3.9k)	westlaw.com (2.0k)	hulu.com (8.9k)
6	xfinity.com (6.9k)	xfinity.com (6.9k)	usps.com (7.1k)	thriftbooks.com (898)	earthlink.net (3.3k)	yelp.com (1.9k)	tiktok.com (8.6k)
7	youtube.com (169k)	cbssports.com (3.9k)	xfinity.com (6.9k)	adp.com (810)	dell.com (3.0k)	activemeasure.com (1.8k)	capitalone.com (7.9k)
8	cbssports.com (3.9k)	dynata.com (22k)	rakuten.com (6.4k)	equitybank.com (723)	zoosk.com (2.4k)	venatusmedia.com (1.8k)	xfinity.com (6.9k)
9	ebay.com (22k)	ebay.com (22k)	wellsfargo.com (5.6k)	priceline.com (663)	ancestry.com (2.1k)	revenueuniverse.com (1.6k)	rakuten.com (6.4k)
10	imdb.com (5.5k)	nielseniq.com (8.5k)	netflix.com (5.2k)	webtoons.com (592)	homedepot.com (2.1k)	kroger.com (1.6k)	washingtonpost.com (5.0k)
11	yougov.com (33k)	weather.com (2.5k)	worldwinner.com (4.6k)	ourfamilywizard.com (562)	kohls.com (1.9k)	spot.im (1.5k)	espn.com (4.9k)
12	nielseniq.com (8.5k)	msn.com (22k)	biggerbooks.com (3.9k)	evergage.com (479)	discover.com (1.9k)	doceree.com (1.5k)	target.com (4.7k)
13	washingtonpost.com (5.0k)	imdb.com (5.5k)	9gag.com (3.7k)	managebuilding.com (473)	venatusmedia.com (1.8k)	attn.tv (1.2k)	bankofamerica.com (4.5k)
14	9gag.com (3.7k)	cnn.com (2.7k)	earthlink.net (3.3k)	overdrive.com (454)	doceree.com (1.5k)	dropbox.com (1.1k)	biggerbooks.com (3.9k)
15	weather.com (2.5k)	twitter.com (89k)	nextdoor.com (3.2k)	meetup.com (447)	verizon.com (1.4k)	pnc.com (1.1k)	hbomax.com (3.3k)
16	cnn.com (2.7k)	youtube.com (169k)	dell.com (3.0k)	narvar.com (337)	tupperware.com (1.3k)	morningjournal.com (1.0k)	citi.com (3.2k)
17	usps.com (7.1k)	kohls.com (1.9k)	iheart.com (2.8k)	coupons.com (336)	attn.tv (1.2k)	ex.co (1.0k)	dell.com (3.0k)
18	rakuten.com (6.4k)	morningjournal.com (1.0k)	zoom.us (2.7k)	udemy.com (313)	phoenix.edu (1.1k)	trulia.com (911)	peacocktv.com (2.8k)
19	hulu.com (8.9k)	9gag.com (3.7k)	airbnb.com (2.6k)	yaysavings.com (313)	prizerebel.com (1.1k)	thriftbooks.com (898)	fedex.com (2.4k)
20	dynata.com (22k)	nytimes.com (3.8k)	zoosk.com (2.4k)	bizpacreview.com (274)	cmix.com (1.1k)	53.com (850)	syf.com (2.3k)
21	kohls.com (1.9k)	zoom.us (2.7k)	creditkarma.com (2.3k)	daisous.com (273)	jcpenney.com (1.0k)	newspapers.com (805)	samsclub.com (2.1k)
22	nytimes.com (3.8k)	chase.com (11k)	ups.com (2.3k)	wootric.com (262)	copart.com (1.0k)	e-rewards.com (764)	zulily.com (2.1k)
23	chase.com (11k)	centurylink.net (1.0k)	ancestry.com (2.1k)	buzzfeed.com (243)	trendmicro.com (979)	kaiserpermanente.org (686)	homedepot.com (2.1k)
24	iheart.com (2.8k)	foxnews.com (2.1k)	samsclub.com (2.1k)	hobbylobby.com (228)	zleague.gg (969)	offerup.com (629)	kohls.com (1.9k)
25	hbomax.com (3.3k)	capitalone.com (7.9k)	homedepot.com (2.1k)	tbdliquids.com (228)	grabpoints.com (882)	qvc.com (603)	discover.com (1.9k)
26	foxnews.com (2.1k)	aol.com (27k)	productreportcard.com (2.1k)	noom.com (215)	box.com (830)	vccs.edu (580)	cbsi.com (1.7k)
27	twitter.com (89k)	dell.com (3.0k)	kohls.com (1.9k)	winred.com (198)	medallia.com (813)	mapquest.com (507)	shein.com (1.6k)
28	zillow.com (18k)	rakuten.com (6.4k)	discover.com (1.9k)	examfx.com (180)	newspapers.com (805)	vimeo.com (503)	adobe.com (1.6k)
29	espn.com (4.9k)	investing.com (670)	activemeasure.com (1.8k)	wgal.com (177)	fidelity.com (789)	ask.com (491)	kroger.com (1.6k)
30	democraticunderground.com (12k)	civicscience.com (3.8k)	venatusmedia.com (1.8k)	fox.com (176)	affirm.com (752)	dynatrace.com (464)	newyorklife.com (1.2k)
31	oregonlive.com (1.3k)	navyfederal.org (1.2k)	duolingo.com (1.7k)	avant.com (173)	wishpond.com (740)	meetup.com (447)	aliexpress.com (1.2k)
32	zoosk.com (2.4k)	paycor.com (1.6k)	cbsi.com (1.7k)	mtsac.edu (171)	equitybank.com (723)	upmc.com (408)	nordstrom.com (1.2k)
33	navyfederal.org (1.2k)	linkedin.com (13k)	experian.com (1.7k)	truegloryhair.com (161)	dominos.com (716)	reserveohio.com (388)	navyfederal.org (1.2k)
34	linkedin.com (13k)	nascar.com (597)	adobe.com (1.6k)	epsilon.com (159)	wurflcloud.com (716)	odysee.com (363)	pnc.com (1.1k)
35	dell.com (3.0k)	google.co.uk (9.4k)	kroger.com (1.6k)	njlottery.com (155)	etrade.com (696)	sutherlandglobal.com (307)	jcpenney.com (1.0k)
36	centurylink.net (1.0k)	hbomax.com (3.3k)	paycor.com (1.6k)	factor75.com (154)	kaiserpermanente.org (686)	bandcamp.com (305)	copart.com (1.0k)
37	capitaloneshopping.com (12k)	zoho.com (10k)	spot.im (1.5k)	higherincomejobs.com (154)	investing.com (670)	forter.com (286)	gap.com (983)
38	hideout.co (8.7k)	adobe.com (1.6k)	honeygain.com (1.5k)	onlygreatjobs.com (149)	bhg.com (660)	twinspires.com (270)	trendmicro.com (979)
39	peacocktv.com (2.8k)	syf.com (2.3k)	verizon.com (1.4k)	reverbnation.com (149)	playsugarhouse.com (652)	freefarmtowngiftshop.com (248)	booking.com (872)
40	msn.com (22k)	spot.im (1.5k)	oregonlive.com (1.3k)	clover.com (144)	adam4adamsfw.com (628)	netspend.com (248)	coursera.org (864)
41	morningjournal.com (1.0k)	huffpost.com (668)	mheducation.com (1.3k)	inmoment.com (143)	veritonic.com (590)	opentable.com (245)	53.com (850)
42	capitalone.com (7.9k)	office.com (13k)	vidyard.com (1.2k)	kmov.com (135)	oldnational.com (563)	partycentersoftware.com (238)	truist.com (822)
43	reddit.com (52k)	webmd.com (554)	newyorklife.com (1.2k)	walmart.com.mx (134)	ourfamilywizard.com (562)	hibid.com (236)	adp.com (810)
44	braincandy.net (772)	wellsfargo.com (5.6k)	navyfederal.org (1.2k)	gerberlife.com (125)	pearson.com (556)	hobbylobby.com (228)	ea.com (803)
45	adobe.com (1.6k)	winnerzcircle.com (5.4k)	attn.tv (1.2k)	yummybazaar.com (119)	gofundme.com (552)	blueconic.net (212)	fidelity.com (789)
46	huffpost.com (668)	venatusmedia.com (1.8k)	shutterfly.com (1.2k)	grabagun.com (104)	slickdeals.net (550)	pinnbank.com (207)	hp.com (768)
47	paycor.com (1.6k)	peacocktv.com (2.8k)	allrecipes.com (1.1k)	gnc.com (100)	adidas.com (538)	boomtrain.com (197)	e-rewards.com (764)
48	discover.com (1.9k)	equibase.com (465)	my-take.com (1.1k)	milbstore.com (98)	emi-rs.com (529)	centercode.com (191)	expedia.com (760)
49	investing.com (670)	cbsnews.com (561)	prizerebel.com (1.1k)	dailyvoice.com (95)	southernliving.com (514)	eyebuydirect.com (185)	tjx.com (750)
50	biggerbooks.com (3.9k)	verizon.com (1.4k)	pnc.com (1.1k)	everyplate.com (92)	geico.com (512)	edx.org (184)	citibankonline.com (691)

Note: This table reports the top 50 domains (rows) contributing to individual-level exposure for each of the seven tracking methods (columns). A domain d's contribution to individual-level exposure is computed as:

$$\text{Contribution}_d^{(s)} = \sum_i \sum_{v \in \mathcal{V}_{id}} |trackers_d^{(s)}|,$$

based on all individual-domain visit instances, weighted by the number of trackers of type s present on domain d. Parentheses report the total number of visits.